Math 105L In Class 2 Practice Solutions

Contents

Limits and Continuity 2

Derivative Function 4

Second Derivative 8

Product and Quotient Rules 10

Inverse Functions and Composition 15

Compound Interest 24

Exponential Functions 26

Logs 29

Chain Rule 33

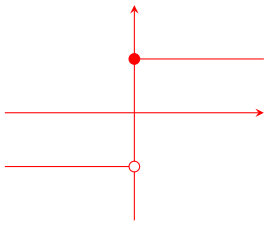
Differentiating Exponentials 39

Implicit Differentiation 43

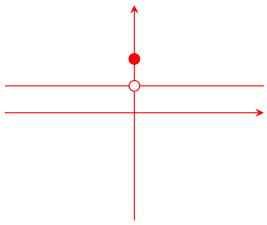
Limits and Continuity

1. Describe the three ways in which a function can fail to be continuous. Draw an example in each case.

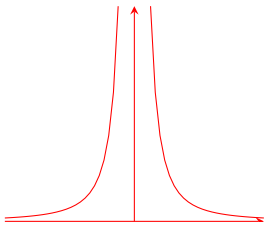
   - We might not be able to know what to expect:



   - We might not get what we expect:



   - What we expect might not be a number:



2. We said that $f(x)$ is *continuous* at $x = a$ if $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) = f(a)$.

   We also said that $f(x)$ is continuous at $x = a$ if we both *know what to expect at* $x = a$ and *get what we expect at* x=a.

   Of the two equalities in the first definition, which says we know what to expect, and which says we get it? Explain.

   $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x)$ says we know what to expect. The two sides approach the same number. That is, $\lim_{x\to a} f(x)$ exists.

   If, *in addition to that*, $\lim_{x\to a} f(x) = f(a)$, we also get what we expect, since the actual value of the function is what we expect it to be.

3. Consider the function $f(x) = \frac{x^2-1}{x+1}$.

   (a) Explain why $f(x)$ is not continuous at $x = -1$.

   $f(-1)$ gives $\frac{0}{0}$, so the function is not defined there. Therefore, regardless of the limits from each side, we cannot get what we expect.

   (b) Consider

$$g(x) = \begin{cases} f(x) & x \neq -1 \\ k & x = -1 \end{cases}$$

   Is there a value of k that makes $g(x)$ continuous? If so, find it. If not, explain why.

   We want $\lim_{x\to -1^+} f(x) = \lim_{x\to -1^-} f(x) = f(-1) = k$:

$$\begin{aligned} \lim_{x\to -1^+} f(x) &= \lim_{x\to -1^+} \frac{x^2-1}{x+1} \\ &= \lim_{x\to -1^+} \frac{(x+1)(x-1)}{x+1} \\ &= \lim_{x\to -1^+} x - 1 = -2 \end{aligned}$$

   (the limit from the negative side is the same).

   So $k = -2$ makes this continuous.

4. Consider the function

$$f(x) = \begin{cases} -1 & x > 0 \\ k & x = 0 \\ 1 & x < 0 \end{cases}$$

   Is there a value of k that makes $f(x)$ continuous? If so, find it. If not, explain why.

   We want $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^-} f(x) = f(0) = k$:

   $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} -1 = -1$

   $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} 1 = 1$

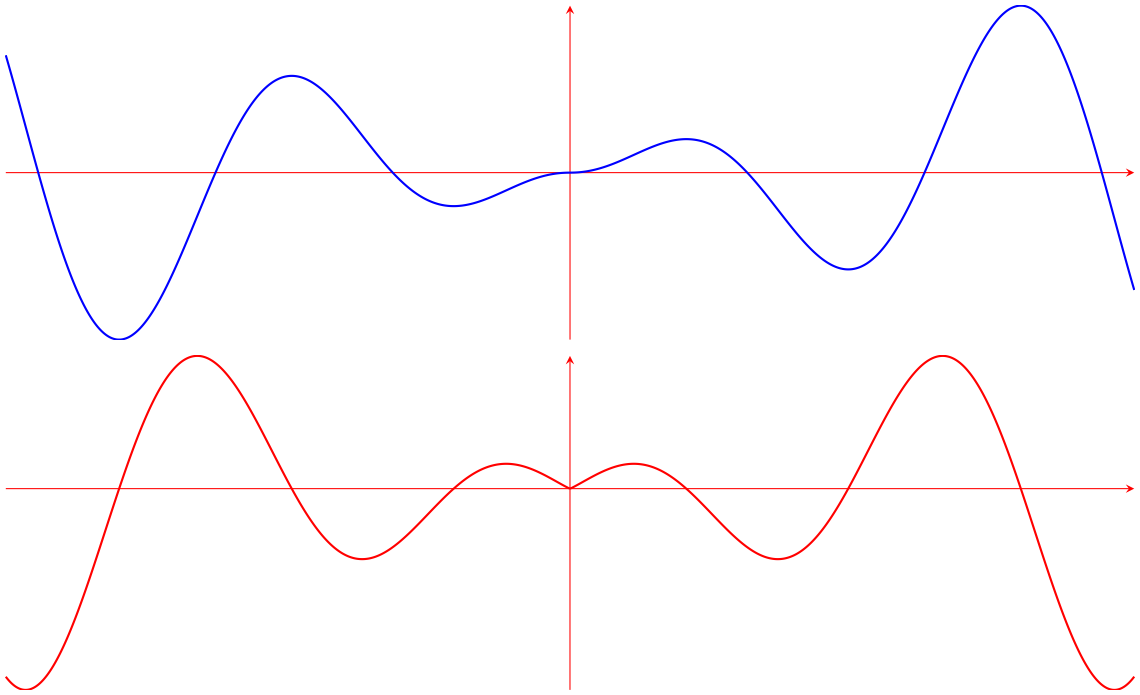
   Since these are not equal, there is no value of k that would make $f(x)$ continuous.

5. Briefly explain why it makes sense that $\lim_{x\to a} kf(x) = k \lim_{x\to a} f(x)$.

   If we know what value we expect $f(x)$ to take at $x = a$ (i.e $\lim_{x\to a} f(x)$ exists), then we expect the function $kf(x)$ to take a value that is k times larger there, since $kf(x)$ just multiplies all y-values by k.

Derivative Function

1. Sketch the graph of the derivative of the following function on the axes below.



2. Answer each of the following questions regarding the relationship between the sign/behavior of a function and its derivative:

   (a) If $f(x)$ is positive for $0 < x < 3$, and negative for $3 < x < 6$, what is the sign of its derivative for $3 < x < 6$? Explain.

   There is no way to know. A function's sign (positive or negative) does not determine whether it is increasing, decreasing, or constant on that range of values. For example, a function can be negative and increasing, negative and decreasing, or both for $3 < x < 6$.

   (b) If $g'(x) > 0$ for all $1 < x < 5$, does this guarantee that $g(x) > 0$ for all $1 < x < 5$? Explain.

   No. The condition $g'(x) > 0$ guarantees that $g(x)$ is strictly increasing for $1 < x < 5$, but it says nothing about the vertical position of the function. For example, $g(x) = x - 10$ has $g'(x) = 1 > 0$ for all x, but $g(x)$ is strictly negative for $1 < x < 5$ (since $g(5) = -5 < 0$).

   (c) Suppose $h(x)$ is continuous and differentiable for $0 < x < 8$. If $h'(x) < 0$ for $0 < x < 8$ and $h(3) = 0$, what can you say about the sign of $h(x)$ for $x < 3$? What about for $x > 3$? Explain.

   Since $h'(x) < 0$, $h(x)$ is strictly decreasing for $0 < x < 8$. Because it equals zero at $x = 3$, values before $x = 3$ must be higher than zero, so $h(x) > 0$ (positive) for $0 < x < 3$. Similarly, values after $x = 3$ must be lower than zero, so $h(x) < 0$ (negative) for $3 < x < 8$.

3. Explain the practical meaning of each of the following statements, including units:

   (a) Suppose that $F(p)$ is a function giving the number of thousands of fish in a county with p ponds in it. Explain what the following quantities mean:

(i) $F(20) = 100$.
(ii) $F'(20) = 6$.

(i) There are 100,000 fish in a county that has 20 ponds.
(ii) When a county has 20 ponds, adding an additional pond increases the total fish population by approximately 6,000 fish (the rate of change is 6,000 fish per pond).

(b) Let $C(w)$ be the monthly cost (in dollars) of heating a home when the average outside temperature during that month is w degrees Fahrenheit (°F). Explain what the following quantities mean:

(i) $C(32) = 210$.
(ii) $C'(32) = -6$.

(i) When the average outside temperature is 32°F, it costs \$210 to heat the home for that month.
(ii) When the average outside temperature is 32°F, each 1°F increase in average outside temperature reduces monthly heating costs by approximately \$6 (the rate of change is −\$6 per °F).

(c) Let $M(t)$ be the total mass of honey (in kilograms) produced by a bee colony t weeks after the start of spring. Explain what the following quantities mean:

(i) $M(8) = 15$.
(ii) $M'(8) = 2.4$.

(i) Eight weeks after the start of spring, the colony has produced a cumulative total of 15 kilograms of honey.
(ii) Eight weeks after the start of spring, the colony is producing honey at a rate of 2.4 kilograms per week.

4. Answer each of the following questions regarding physical interpretations of functions and rates of change:

(a) Let $A(t)$ be the depth below sea level of a submarine t hours after its launch, in feet.

(i) Explain why the statement $A(5) = -1000$ makes no physical sense. Include units in your answer.
(ii) Suppose that $A'(5) = -500$. Discuss the motion of the submarine at this time, including direction, speed, and units.

(i) Since $A(t)$ measures depth below sea level in feet, $A(5) = -1000$ would mean the submarine is 1,000 feet *above* sea level (flying in the air), which is nonsensical for a submarine.
(ii) Since $A'(5) = -500$ ft/hr, the depth is decreasing at 500 feet per hour. This means the submarine is ascending towards the surface at a speed of 500 ft/hr.

(b) Let $H(t)$ be the altitude above ground level of a weather balloon, in meters, t minutes after release.

(i) Explain why the statement $H(12) = -80$ makes no physical sense. Include units.
(ii) Suppose that $H'(12) = -5$. Discuss the motion of the balloon at this time, including direction, speed, and units.

(i) Since $H(t)$ measures height above the ground in meters, $H(12) = -80$ would mean the balloon is 80 meters underground, which is physically impossible.

(ii) Since $H'(12) = -5$ m/min, altitude is decreasing at 5 meters per minute. The balloon is descending towards the ground at a speed of 5 meters per minute.

(c) Let $V(t)$ be the volume of water (in gallons) remaining in an agricultural storage reservoir t days after the start of a drought. (Assume the reservoir cannot have a negative volume.)

(i) Explain why the statement $V(40) = -500$ makes no practical sense. Include units.

(ii) Suppose that $V'(40) = -120$. Discuss what is happening to the reservoir at this time, including rate and units.

(i) A physical container cannot hold a negative volume of water; once empty, the volume remains at 0 gallons. A value of -500 gallons is physically impossible.

(ii) Since $V'(40) = -120$ gal/day, the volume of water in the reservoir is decreasing at a rate of 120 gallons per day (due to evaporation and consumption).

5. For each of the following functions, compute the derivative $f'(x)$ *from the definition of the derivative, without using any differentiation rules* ($f'(x) = \lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$):

(a) $f(x) = x^2 + 2x$

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{((x+h)^2 + 2(x+h)) - (x^2+2x)}{h} \\ &= \lim_{h\to 0} \frac{x^2 + 2xh + h^2 + 2x + 2h - x^2 - 2x}{h} \\ &= \lim_{h\to 0} \frac{2xh + h^2 + 2h}{h} \\ &= \lim_{h\to 0} \frac{h(2x + h + 2)}{h} \\ &= \lim_{h\to 0} (2x + h + 2) = 2x + 2. \end{aligned}$$

(b) $f(x) = 3x^2 - 5x + 1$

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{(3(x+h)^2 - 5(x+h) + 1) - (3x^2 - 5x + 1)}{h} \\ &= \lim_{h\to 0} \frac{3(x^2 + 2xh + h^2) - 5x - 5h + 1 - 3x^2 + 5x - 1}{h} \\ &= \lim_{h\to 0} \frac{3x^2 + 6xh + 3h^2 - 5h - 3x^2}{h} \\ &= \lim_{h\to 0} \frac{6xh + 3h^2 - 5h}{h} \\ &= \lim_{h\to 0} \frac{h(6x + 3h - 5)}{h} \\ &= \lim_{h\to 0} (6x + 3h - 5) = 6x - 5. \end{aligned}$$

(c) $f(x) = \frac{1}{x+2}$

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{\frac{1}{(x+h)+2} - \frac{1}{x+2}}{h} \\ &= \lim_{h\to 0} \frac{1}{h}\left[\frac{(x+2)-(x+h+2)}{(x+h+2)(x+2)}\right] \\ &= \lim_{h\to 0} \frac{1}{h}\left[\frac{-h}{(x+h+2)(x+2)}\right] \\ &= \lim_{h\to 0} \frac{-1}{(x+h+2)(x+2)} = -\frac{1}{(x+2)^2}. \end{aligned}$$

Second Derivative

1. If $f''(x) > 0$, then which of the following is true? Explain why each is true, false, or unknowable.

   (a) The function $f(x)$ is positive.

   (b) The tangent lines to $f(x)$ are increasing.

   (c) The slopes of the tangent lines to $f(x)$ are increasing.

   (a) and (b) are both unknowable. The function may or may not be positive. The tangent lines could be increasing ($f'(x) > 0$) or decreasing ($f'(x) < 0$), but we have no idea from this.

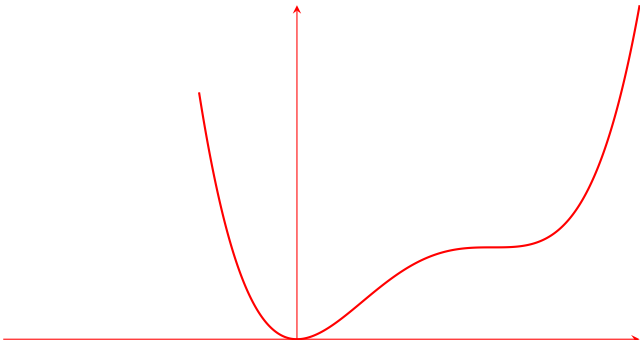
   (c) is true. $f''(x) > 0$ means exactly that: as x increases, the slopes of the tangent lines increase too, since $f''(x)$ is the rate of change of the slopes.

2. Sketch graphs of smooth continuous functions with domain all real numbers matching each one of the following descriptions, or explain why it's impossible.

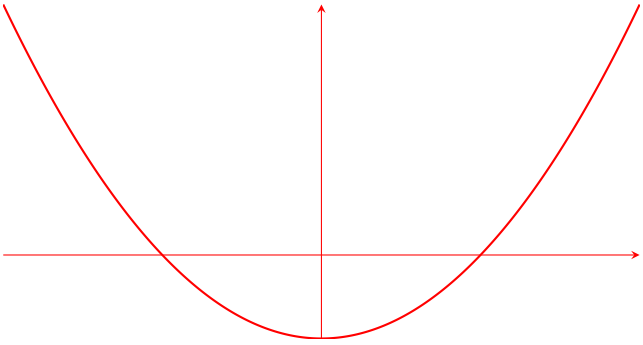
   (a) $f(x) > 0$, $f'(x) < 0$, $f''(x) < 0$, and $\lim_{x\to\infty} = 5$.

   Impossible. If the function is decreasing and concave down, it gets steeper as x increases, meaning it can't have a HA at $y = 5$.

   (b) $f(x)$ has two inflection points, $f'(x) < 0$ for $x < 0$, $f'(x) \geq 0$ for $x > 0$.



   (c) $f(x)$ has no inflection points, $f'(x) = 0$ at exactly one point, $f(x)$ has two zeros.



3. True or False? Explain your answers.

   (a) If a function is concave up, then $f''(x) > 0$.

   False. $f(x) = x^4$ is concave up, but has $f''(x) = 12x^2$, which is 0 at $x = 0$.

(b) If a function is concave down, then $f''(x) \leq 0$.

True. If at any point, $f''(x) > 0$, $f(x)$ is concave up there, so if $f(x)$ is concave down, $f''(x)$ must be non-positive.

(c) When a function turns around, it has an inflection point.

False. Turning points and inflection points are not the same thing.

(d) It is possible for a function to be decreasing, but have increasing slopes.

True. This just means it's decreasing and concave up. So for example, e^{-x} for all real numbers, or x^2 for $x < 0$.

4. Fill in the blanks: If a function is increasing, then its derivative or slope is positive. If the slopes of a function are decreasing, then its second derivative is negative.

5. The position of an object above the ground at time t is given by a function $s(t)$, where $s'(t) < 0$ and $s''(t) > 0$. Is the object rising or falling? Is it speeding up or slowing down? Explain.

The object is falling, since its height is decreasing ($s'(t) < 0$).

The object is slowing down: its velocity is negative, but increasing (since $s''(t) > 0$, and $s''(t)$ is the rate of change of $s'(t)$, the velocity). So its velocity is becoming less negative, which is closer to zero. Therefore the object is slowing down.

Product and Quotient Rules

1. Differentiate $f(x) = x^2(3x^3 + 1)$.

   Use the product rule. Let $u(x) = x^2$, $v(x) = 3x^3 + 1$.

   $$u'(x) = 2x, \quad v'(x) = 9x^2$$

   $$f'(x) = u'v + uv' = 2x(3x^3 + 1) + x^2(9x^2)$$

2. Differentiate $g(x) = (x^3 - 2x)(x + 5)$.

   Product rule: $u(x) = x^3 - 2x$, $v(x) = x + 5$.

   $$u'(x) = 3x^2 - 2, \quad v'(x) = 1$$

   $$g'(x) = u'v + uv' = (3x^2 - 2)(x + 5) + (x^3 - 2x)(1)$$

3. Differentiate $h(x) = \sqrt{x}\,(2x^2 - 1)$. (Assume $x > 0$.)

   Write $\sqrt{x} = x^{1/2}$. Let $u(x) = x^{1/2}$, $v(x) = 2x^2 - 1$.

   $$u'(x) = \tfrac{1}{2}x^{-1/2}, \quad v'(x) = 4x$$

   $$h'(x) = u'v + uv' = \tfrac{1}{2}x^{-1/2}(2x^2 - 1) + x^{1/2}(4x)$$

4. Differentiate $p(x) = (x - 1)^2(x^2 + 2)$.

   Let $u(x) = (x - 1)^2$, $v(x) = x^2 + 2$.

   $$u'(x) = 2(x - 1), \quad v'(x) = 2x$$

   $$p'(x) = u'v + uv' = 2(x - 1)(x^2 + 2) + (x - 1)^2(2x)$$

5. Differentiate $q(x) = e^x(x^2 - 3x + 1)$.

   Let $u(x) = e^x$, $v(x) = x^2 - 3x + 1$.

   $$u'(x) = e^x, \quad v'(x) = 2x - 3$$

   $$q'(x) = u'v + uv' = e^x(x^2 - 3x + 1) + e^x(2x - 3)$$

6. Differentiate $r(x) = \dfrac{x^2 + 1}{x}$.

   Quotient rule: $u(x) = x^2 + 1$, $v(x) = x$.

   $$u'(x) = 2x, \quad v'(x) = 1$$

   $$r'(x) = \frac{u'v - uv'}{v^2} = \frac{2x(x) - (x^2 + 1)(1)}{x^2}$$

7. Differentiate $s(x) = \dfrac{3x-1}{x^2+1}$.

Let $u(x) = 3x-1$, $v(x) = x^2+1$.

$$u'(x) = 3, \quad v'(x) = 2x$$

$$s'(x) = \frac{u'v - uv'}{v^2} = \frac{3(x^2+1) - (3x-1)(2x)}{(x^2+1)^2}$$

8. Differentiate $t(x) = \dfrac{x^3+2}{x-1}$.

Quotient rule: $u(x) = x^3+2$, $v(x) = x-1$.

$$u'(x) = 3x^2, \quad v'(x) = 1$$

$$t'(x) = \frac{u'v - uv'}{v^2} = \frac{3x^2(x-1) - (x^3+2)(1)}{(x-1)^2}$$

9. Differentiate $u(x) = \dfrac{\sqrt{x}}{x+2}$.

Write $\sqrt{x} = x^{1/2}$. Let $p(x) = x^{1/2}$, $q(x) = x+2$.

$$p'(x) = \tfrac{1}{2}x^{-1/2}, \quad q'(x) = 1$$

$$u'(x) = \frac{p'q - pq'}{q^2} = \frac{\frac{1}{2}x^{-1/2}(x+2) - x^{1/2}(1)}{(x+2)^2}$$

10. Differentiate $v(x) = (x^2+1)(2x^3-x)$.

Let $a(x) = x^2+1$, $b(x) = 2x^3-x$.

$$a'(x) = 2x, \quad b'(x) = 6x^2-1$$

$$v'(x) = a'b + ab' = 2x(2x^3-x) + (x^2+1)(6x^2-1)$$

11. Find the tangent line to $f(x) = x^2(3x+1)$ at $x = 1$.

Use the product rule. Let $u(x) = x^2$, $v(x) = 3x+1$.

$$u'(x) = 2x, \quad v'(x) = 3$$

$$f'(x) = u'v + uv' = 2x(3x+1) + x^2(3)$$

Evaluate at $x = 1$:

$$f(1) = 1^2(3\cdot 1+1) = 4, \quad f'(1) = 11$$

Point-slope form:

$$y - f(1) = f'(1)(x-1) \quad \text{so} \quad y - 4 = 11(x-1)$$

Final tangent line:

$$y = 4 + 11(x-1)$$

12. Find the tangent line to $g(x) = (x^3 - 2x)(x + 2)$ at $x = 0$.

Product rule. Let $u(x) = x^3 - 2x$, $v(x) = x + 2$.

$$u'(x) = 3x^2 - 2, \quad v'(x) = 1$$

$$g'(x) = u'v + uv' = (3x^2 - 2)(x + 2) + (x^3 - 2x)(1)$$

At $x = 0$:

$$g(0) = 0, \quad g'(0) = -4$$

Point-slope form:

$$y - g(0) = g'(0)(x - 0) \quad \text{so} \quad y - 0 = -4(x - 0)$$

Final tangent line:

$$y = -4x$$

13. Find the tangent line to $h(x) = \sqrt{x}(2x^2 - 1)$ at $x = 1$.

Write $\sqrt{x} = x^{1/2}$. Let $u(x) = x^{1/2}$, $v(x) = 2x^2 - 1$.

$$u'(x) = \frac{1}{2}x^{-1/2}, \quad v'(x) = 4x$$

$$h'(x) = u'v + uv' = \frac{1}{2}x^{-1/2}(2x^2 - 1) + x^{1/2}(4x)$$

At $x = 1$:

$$h(1) = 1, \quad h'(1) = \frac{9}{2}$$

Point-slope form:

$$y - h(1) = h'(1)(x - 1) \quad \text{so} \quad y - 1 = \frac{9}{2}(x - 1)$$

Final tangent line:

$$y = 1 + \frac{9}{2}(x - 1)$$

14. Find the tangent line to $p(x) = (x - 1)^2(x^2 + 1)$ at $x = 2$.

Let $u(x) = (x - 1)^2$, $v(x) = x^2 + 1$.

$$u'(x) = 2(x - 1), \quad v'(x) = 2x$$

$$p'(x) = u'v + uv' = 2(x - 1)(x^2 + 1) + (x - 1)^2(2x)$$

At $x = 2$:

$$p(2) = 5, \quad p'(2) = 14$$

Point-slope form:

$$y - p(2) = p'(2)(x - 2) \quad \text{so} \quad y - 5 = 14(x - 2)$$

Final tangent line:

$$y = 5 + 14(x - 2)$$

15. Find the tangent line to $q(x) = e^x(x^2 - 2x)$ at $x = 0$.

Let $u(x) = e^x$, $v(x) = x^2 - 2x$.

$$u'(x) = e^x, \quad v'(x) = 2x - 2$$

$$q'(x) = u'v + uv' = e^x(x^2 - 2x) + e^x(2x - 2)$$

At $x = 0$:

$$q(0) = 0, \quad q'(0) = -2$$

Point-slope form:

$$y - q(0) = q'(0)(x - 0) \quad \text{so} \quad y - 0 = -2(x - 0)$$

Final tangent line:

$$y = -2x$$

16. Find the tangent line to $r(x) = \frac{x^2+1}{x}$ at $x = 1$.

Quotient rule. Let $u(x) = x^2 + 1$, $v(x) = x$.

$$u'(x) = 2x, \quad v'(x) = 1$$

$$r'(x) = \frac{u'v - uv'}{v^2} = \frac{2x \cdot x - (x^2 + 1)}{x^2}$$

At $x = 1$:

$$r(1) = 2, \quad r'(1) = 0$$

Point-slope form:

$$y - r(1) = r'(1)(x - 1) \quad \text{so} \quad y - 2 = 0 \cdot (x - 1)$$

Final tangent line:

$$y = 2$$

17. Find the tangent line to $s(x) = \frac{3x-1}{x^2+1}$ at $x = 1$.

Let $u(x) = 3x - 1$, $v(x) = x^2 + 1$.

$$u'(x) = 3, \quad v'(x) = 2x$$

$$s'(x) = \frac{u'v - uv'}{v^2} = \frac{3(x^2 + 1) - (3x - 1)(2x)}{(x^2 + 1)^2}$$

At $x = 1$:

$$s(1) = 1, \quad s'(1) = \frac{1}{2}$$

Point-slope form:

$$y - s(1) = s'(1)(x - 1) \quad \text{so} \quad y - 1 = \frac{1}{2}(x - 1)$$

Final tangent line:

$$y = 1 + \frac{1}{2}(x - 1)$$

18. Find the tangent line to $t(x) = \frac{x^3+1}{x-1}$ at $x = 2$.

Let $u(x) = x^3 + 1$, $v(x) = x - 1$.

$$u'(x) = 3x^2, \quad v'(x) = 1$$

$$t'(x) = \frac{u'v - uv'}{v^2} = \frac{3x^2(x - 1) - (x^3 + 1)}{(x - 1)^2}$$

At $x = 2$:

$$t(2) = 9, \quad t'(2) = 3$$

Point-slope form:

$$y - t(2) = t'(2)(x - 2) \quad \text{so} \quad y - 9 = 3(x - 2)$$

Final tangent line:

$$y = 9 + 3(x - 2)$$

19. Find the tangent line to $u(x) = \frac{\sqrt{x}}{x+1}$ at $x = 4$.

Write $\sqrt{x} = x^{1/2}$. Let $p(x) = x^{1/2}$, $q(x) = x + 1$.

$$p'(x) = \frac{1}{2}x^{-1/2}, \quad q'(x) = 1$$

$$u'(x) = \frac{p'q - pq'}{q^2} = \frac{\frac{1}{2}x^{-1/2}(x+1) - x^{1/2}}{(x+1)^2}$$

At $x = 4$:

$$u(4) = \frac{2}{5}, \quad u'(4) = -\frac{3}{100}$$

Point-slope form:

$$y - u(4) = u'(4)(x - 4) \quad \text{so} \quad y - \frac{2}{5} = -\frac{3}{100}(x - 4)$$

Final tangent line:

$$y = \frac{2}{5} - \frac{3}{100}(x - 4)$$

20. Find the tangent line to $v(x) = (x^2 + 1)(2x^2 - x)$ at $x = -1$.

Let $a(x) = x^2 + 1$, $b(x) = 2x^2 - x$.

$$a'(x) = 2x, \quad b'(x) = 4x - 1$$

$$v'(x) = a'b + ab' = 2x(2x^2 - x) + (x^2 + 1)(4x - 1)$$

At $x = -1$:

$$v(-1) = 6, \quad v'(-1) = -16$$

Point-slope form:

$$y - v(-1) = v'(-1)(x + 1) \quad \text{so} \quad y - 6 = -16(x + 1)$$

Final tangent line:

$$y = 6 - 16(x + 1)$$

Inverse Functions and Composition

1. Explain the meaning of each of the following statements in practical terms:

   (a) Let $P(x)$ be the number of elephants within a distance of x meters from my bathtub. Explain the meaning of each of the following in practical terms, including units:

   (i) $P(20) = 4$
   (ii) $P^{-1}(10) = 55$
   (iii) $P^{-1}(0) = 8$

   (i) There are 4 elephants within a distance of 20 meters from my bathtub.
   (ii) A search radius of 55 meters from my bathtub contains 10 elephants.
   (iii) There are 0 elephants within 8 meters of my bathtub (i.e., the closest elephant is at least 8 meters away).

   (b) A factory produces custom bicycles. Let $C(n)$ be the total cost (in dollars) of producing n bicycles in a day, and let $N(t)$ be the number of bicycles produced after the factory has been running for t hours. Explain the meaning of each of the following in practical terms:

   (i) $C(50) = 15{,}000$
   (ii) $C^{-1}(8000) = 25$
   (iii) The composite function $K(t) = C(N(t))$
   (iv) $K^{-1}(12{,}000) = 4$
   (v) Explain why the expression $N(C(50))$ does not make physical sense.

   (i) It costs \$15,000 to produce 50 bicycles in a day.
   (ii) A budget of \$8,000 allows the factory to produce 25 bicycles.
   (iii) Represents the total cost (in dollars) of production after operating the factory for t hours.
   (iv) It takes 4 hours of operation for the production cost to reach \$12,000.
   (v) $N(C(50))$ does not make sense because $C(50)$ outputs cost in dollars, whereas $N(t)$ requires time in hours as its input.

   (c) Let $T(h)$ be the temperature (in °C) of the atmosphere at an altitude of h kilometers above sea level, and let $D(T)$ be the density of air (in kg/m^3) when the air temperature is T°C. Explain the meaning of each of the following, including units:

   (i) $T(4) = -11$
   (ii) $T^{-1}(0) = 2.5$
   (iii) The composite function $\rho(h) = D(T(h))$
   (iv) $\rho^{-1}(0.9) = 3$

   (i) At an altitude of 4 kilometers above sea level, the atmospheric temperature is -11°C.
   (ii) The altitude where the temperature is 0°C is 2.5 kilometers above sea level (the freezing level is at 2.5 km).
   (iii) Gives the density of air (in kg/m^3) at an altitude of h kilometers above sea level.
   (iv) The air density is 0.9 kg/m^3 at an altitude of 3 kilometers above sea level.

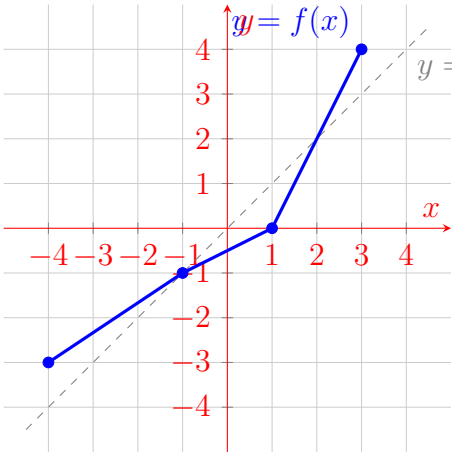
(d) A patient is administered a medication. Let $C(t)$ be the concentration of the drug in the bloodstream (in mg/L) t hours after administration, and let $R(c)$ be the patient's pain relief score (on a scale from 0 to 100) at drug concentration c mg/L. Explain the meaning of each of the following in practical terms:

(i) $C(2) = 18$
(ii) $C^{-1}(5) = 9$
(iii) The composite function $S(t) = R(C(t))$
(iv) $S^{-1}(80) = 3$

(i) Two hours after taking the medication, the drug concentration in the bloodstream is 18 mg/L.
(ii) It takes 9 hours after administration for the drug concentration to decrease to 5 mg/L.
(iii) Gives the patient's pain relief score (from 0 to 100) t hours after taking the medication.
(iv) The patient's pain relief score reaches 80 points 3 hours after taking the medication.

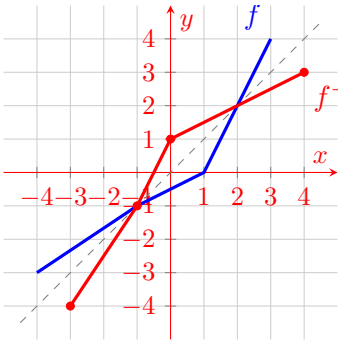
2. For each of the following functions $y = f(x)$ shown on the axes below, sketch the graph of the inverse function $y = f^{-1}(x)$ and answer the associated questions:

(a) The graph of $y = f(x)$ is shown below, with domain $-4 \leq x \leq 3$ and range $-3 \leq y \leq 4$.



(i) On the axes above, sketch the graph of the inverse function $y = f^{-1}(x)$.
(ii) Given that the domain of $f(x)$ is $-4 \leq x \leq 3$ and its range is $-3 \leq y \leq 4$, state the domain and range of $f^{-1}(x)$.
(iii) At what point do the graphs of $f(x)$ and $f^{-1}(x)$ intersect?

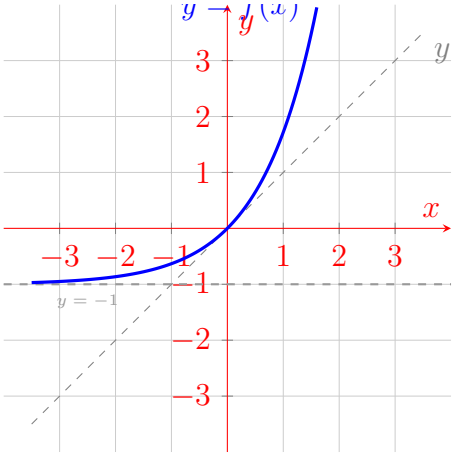
(i) Reflecting each vertex (a, b) across $y = x$ to (b, a) yields $(-3, -4)$, $(-1, -1)$, $(0, 1)$, and $(4, 3)$. Connecting these vertices with line segments gives the graph of $f^{-1}(x)$ (shown below in red):



(ii) Swapping the domain and range of f: the domain of $f^{-1}(x)$ is $-3 \le x \le 4$ and its range is $-4 \le y \le 3$.

(iii) They intersect at $(-1, -1)$ on the line $y = x$.

(b) The graph of $f(x) = e^x - 1$ is shown below, with domain $-\infty < x < \infty$ and range $y > -1$. The horizontal asymptote at $y = -1$ is indicated by a dashed line.



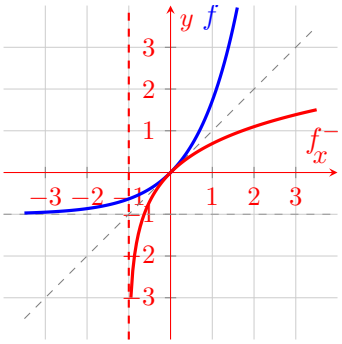
(i) What asymptote does the inverse function $y = f^{-1}(x)$ have? Write its equation.

(ii) On the axes above, sketch the graph of $y = f^{-1}(x)$.

(iii) Given that the domain of $f(x)$ is $-\infty < x < \infty$ and its range is $y > -1$, state the domain and range of $f^{-1}(x)$.

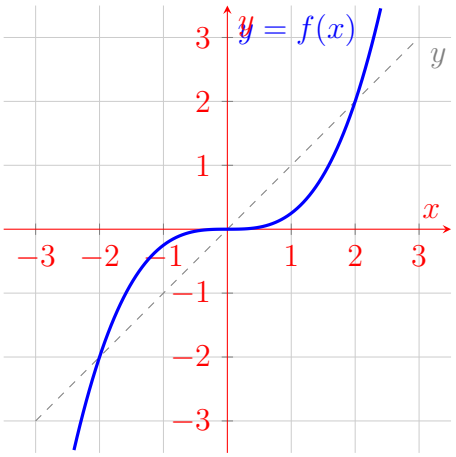
(i) The horizontal asymptote $y = -1$ reflects into a *vertical* asymptote at $x = -1$.

(ii) The inverse is $f^{-1}(x) = \ln(x + 1)$ (shown below in red):



(iii) Swapping domain and range: the domain of $f^{-1}(x)$ is $x > -1$ and its range is $-\infty < y < \infty$.

(c) The graph of $f(x) = \frac{1}{4}x^3$ is shown below along with the diagonal line $y = x$, with domain $-\infty < x < \infty$ and range $-\infty < y < \infty$.

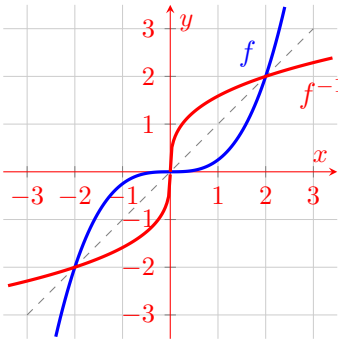


(i) On the axes above, sketch the graph of the inverse function $y = f^{-1}(x)$.

(ii) Given that the domain of $f(x)$ is $-\infty < x < \infty$ and its range is $-\infty < y < \infty$, state the domain and range of $f^{-1}(x)$.

(iii) The graph of $f(x)$ has a horizontal tangent line at the origin $(0, 0)$. What is the tangent line to the graph of $f^{-1}(x)$ at the origin? Explain geometrically.

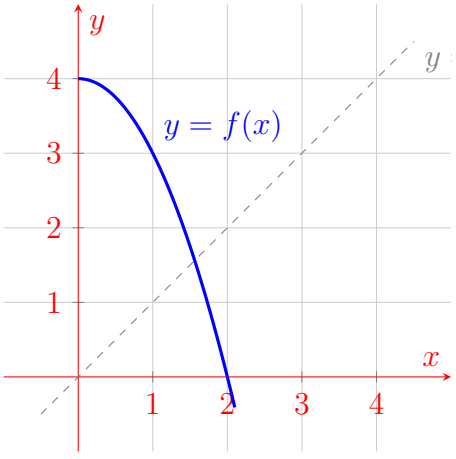
(i) The inverse function is $f^{-1}(x) = \sqrt[3]{4x}$, which passes through $(-2, -2)$, $(0, 0)$, and $(2, 2)$ (shown in red below):



(ii) The domain of $f^{-1}(x)$ is $-\infty < x < \infty$ and its range is $-\infty < y < \infty$.

(iii) The tangent line to $f^{-1}(x)$ at $(0, 0)$ is *vertical* ($x = 0$), because reflection across $y = x$ swaps the horizontal tangent (slope 0) into a vertical line (undefined slope).

(d) The graph below shows the function $f(x) = 4 - x^2$ defined on the restricted domain $x \geq 0$, where its range is $y \leq 4$.



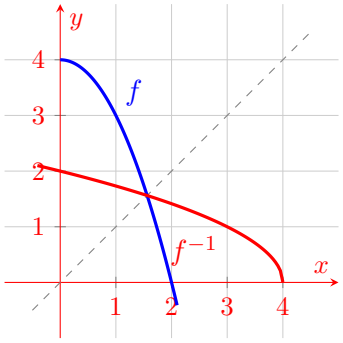
(i) Explain why $f(x) = 4 - x^2$ on $-\infty < x < \infty$ is not invertible, but becomes invertible when its domain is restricted to $x \geq 0$.

(ii) On the axes above, sketch the graph of $y = f^{-1}(x)$.

(iii) Given that the domain of this restricted function $f(x)$ is $x \geq 0$ and its range is $y \leq 4$, state the domain and range of $f^{-1}(x)$.

(i) On $-\infty < x < \infty$, the parabola fails the Horizontal Line Test because every positive y-value comes from two different x-values ($x = \pm\sqrt{4-y}$). Restricting to $x \geq 0$ makes the function strictly decreasing and one-to-one, so it is invertible.

(ii) The inverse is $f^{-1}(x) = \sqrt{4-x}$ for $x \leq 4$ (shown below in red):

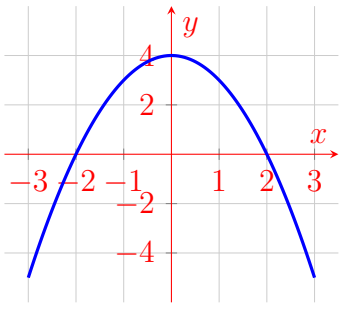


(iii) Swapping the domain and range of f: the domain of $f^{-1}(x)$ is $x \leq 4$ and its range is $y \geq 0$.

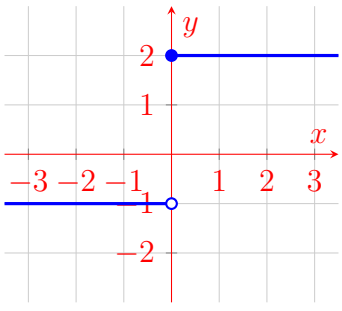
3. For each pair of functions f and g shown below, sketch the graphs of the composite functions $y = f(g(x))$ and $y = g(f(x))$:

(a) The graphs of $f(x) = 4 - x^2$ and $g(x) = \begin{cases} -1 & x < 0 \\ 2 & x \geq 0 \end{cases}$ are given below.

$y = f(x)$



$y = g(x)$

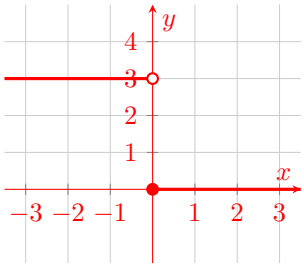


(i) Sketch the graph of $y = f(g(x))$. Explain your reasoning.

(ii) Sketch the graph of $y = g(f(x))$. Explain your reasoning.

(i) For $x < 0$, $g(x) = -1 \implies f(g(x)) = f(-1) = 3$. For $x \geq 0$, $g(x) = 2 \implies f(g(x)) = f(2) = 0$.

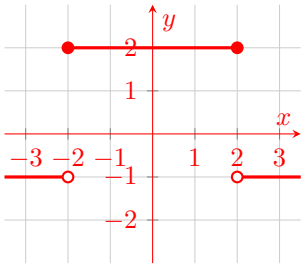
$y = f(g(x))$



(ii) $g(u) = 2$ when $u \geq 0$, and -1 when $u < 0$. Since $u = f(x) = 4 - x^2 \geq 0$ for $-2 \leq x \leq 2$:

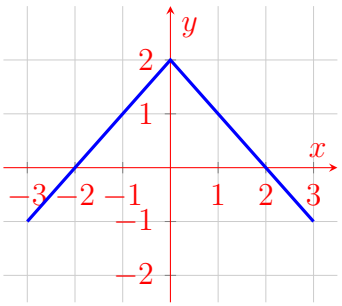
$$g(f(x)) = \begin{cases} 2 & \text{if } -2 \leq x \leq 2 \\ -1 & \text{if } x < -2 \text{ or } x > 2 \end{cases}$$

$y = g(f(x))$

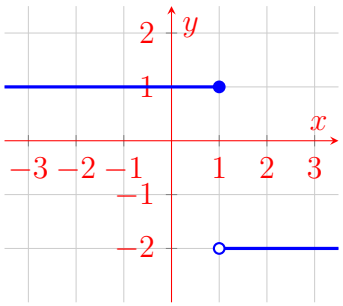


(b) The graphs of $f(x) = 2 - |x|$ and $g(x) = \begin{cases} 1 & x \leq 1 \\ -2 & x > 1 \end{cases}$ are shown below.

$y = f(x)$



$y = g(x)$

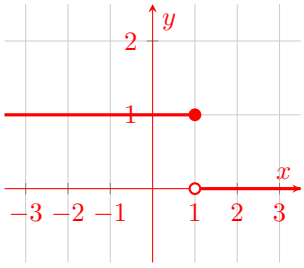


(i) Sketch the graph of $y = f(g(x))$. Explain your reasoning.

(ii) Sketch the graph of $y = g(f(x))$. Explain your reasoning.

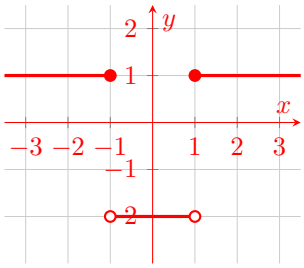
(i) For $x \leq 1$, $g(x) = 1 \implies f(g(x)) = f(1) = 1$. For $x > 1$, $g(x) = -2 \implies f(g(x)) = f(-2) = 0$.

$y = f(g(x))$



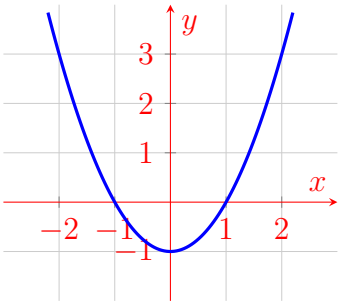
(ii) $g(u) = 1$ when $u \le 1$, and -2 when $u > 1$. Here $u = f(x) = 2 - |x| > 1 \iff |x| < 1 \iff -1 < x < 1$. Therefore, $g(f(x)) = -2$ for $-1 < x < 1$, and 1 for $x \le -1$ or $x \ge 1$.

$y = g(f(x))$

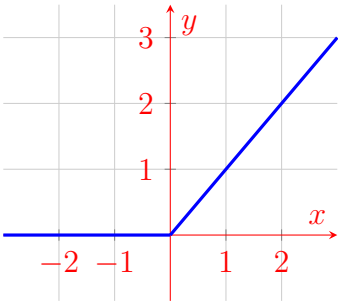


(c) **[Challenge]** The graphs of $f(x) = x^2 - 1$ and the ramp function $g(x) = \begin{cases} 0 & x < 0 \\ x & x \ge 0 \end{cases}$ are given below.

$y = f(x)$



$y = g(x)$



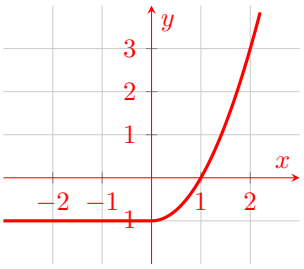
(i) Write a piecewise formula for $f(g(x))$, and sketch its graph.

(ii) Write a piecewise formula for $g(f(x))$, and sketch its graph.

(i) For $x < 0$, $g(x) = 0 \implies f(g(x)) = f(0) = -1$. For $x \ge 0$, $g(x) = x \implies f(g(x)) = f(x) = x^2 - 1$.

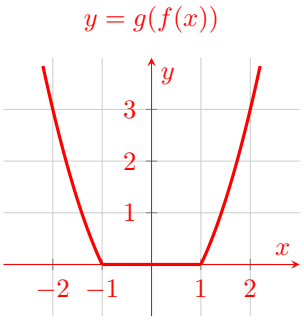
$$f(g(x)) = \begin{cases} -1 & \text{if } x < 0 \\ x^2 - 1 & \text{if } x \ge 0 \end{cases}$$

$y = f(g(x))$



(ii) $g(u) = 0$ when $u < 0$, and u when $u \ge 0$. Here $u = f(x) = x^2 - 1 < 0$ when $-1 < x < 1$, and ≥ 0 when $|x| \ge 1$.

$$g(f(x)) = \begin{cases} 0 & \text{if } -1 < x < 1 \\ x^2 - 1 & \text{if } x \le -1 \text{ or } x \ge 1 \end{cases}$$



4. For each of the following functions, find a formula for the inverse function and state its domain and range:

   (a) Consider the rational function $f(x) = \frac{3x+2}{x-4}$, which has domain $x \neq 4$ and range $y \neq 3$.

   (i) Find a formula for $f^{-1}(x)$. Show all your algebraic steps clearly.

   (ii) State the domain and range of $f^{-1}(x)$.

   (i) Set $y = \frac{3x+2}{x-4}$ and solve for x:

$$y(x-4) = 3x+2 \implies yx - 4y = 3x+2 \implies x(y-3) = 4y+2 \implies x = \frac{4y+2}{y-3}.$$

   Hence:

$$f^{-1}(x) = \frac{4x+2}{x-3}.$$

   (ii) The domain of $f^{-1}(x)$ is $x \neq 3$ and its range is $y \neq 4$.

   (b) Consider the exponential function $g(x) = 5 - 2e^{3x-1}$, which has domain $-\infty < x < \infty$ and range $y < 5$.

   (i) Find a formula for $g^{-1}(x)$. Show all your steps.

   (ii) State the domain and range of $g^{-1}(x)$.

   (i) Set $y = 5 - 2e^{3x-1}$ and isolate x:

$$2e^{3x-1} = 5 - y \implies e^{3x-1} = \frac{5-y}{2}.$$

   Taking the natural logarithm:

$$3x - 1 = \ln\left(\frac{5-y}{2}\right) \implies x = \frac{1 + \ln\left(\frac{5-y}{2}\right)}{3}.$$

   Hence:

$$g^{-1}(x) = \frac{1 + \ln\left(\frac{5-x}{2}\right)}{3} = \frac{1 + \ln(5-x) - \ln 2}{3}.$$

   (ii) The domain of $g^{-1}(x)$ is $x < 5$ and its range is $-\infty < y < \infty$.

   (c) Consider the logarithmic function $h(x) = 4\ln(2x-6)+1$, which has domain $x > 3$ and range $-\infty < y < \infty$.

   (i) Find a formula for $h^{-1}(x)$, showing all your work.

   (ii) State the domain and range of $h^{-1}(x)$.

(i) Set $y = 4\ln(2x-6)+1$ and solve for x:

$$y-1 = 4\ln(2x-6) \implies \ln(2x-6) = \frac{y-1}{4}.$$

Exponentiating base e:

$$2x-6 = e^{\frac{y-1}{4}} \implies x = 3+\frac{1}{2}e^{\frac{y-1}{4}}.$$

Hence:

$$h^{-1}(x) = 3+\frac{1}{2}e^{\frac{x-1}{4}}.$$

(ii) The domain of $h^{-1}(x)$ is $-\infty < x < \infty$ and its range is $y > 3$.

(d) Consider the base-10 exponential function $k(x) = 2\cdot 10^{3x} - 4$, which has domain $-\infty < x < \infty$ and range $y > -4$.

(i) Find a formula for $k^{-1}(x)$. Show all your steps.

(ii) State the domain and range of $k^{-1}(x)$.

(i) Set $y = 2\cdot 10^{3x} - 4$ and solve for x:

$$y+4 = 2\cdot 10^{3x} \implies 10^{3x} = \frac{y+4}{2}.$$

Taking the base-10 logarithm:

$$3x = \log_{10}\left(\frac{y+4}{2}\right) \implies x = \frac{1}{3}\log_{10}\left(\frac{y+4}{2}\right).$$

Hence:

$$k^{-1}(x) = \frac{1}{3}\log_{10}\left(\frac{x+4}{2}\right).$$

(ii) The domain of $k^{-1}(x)$ is $x > -4$ and its range is $-\infty < y < \infty$.

Compound Interest

1. Which of the following is good approximation of $Ae^{0.04t}$? Explain your answer.

   (a) $A\left(1+\frac{1}{0.04}\right)^{1,000,000}$

   (b) $A\left(1+\frac{0.04}{1,000,000}\right)^{1,000,000}$

   (c) $A\left(1+\frac{1,000,0000}{0.04}\right)^{1,000,000}$

   (d) $A\left(1+\frac{1}{0.04}\right)^{0.04}$

   (b) is the right answer. We know that $Ae^{0.04t}$ is the amount of money you have if you invest $\$A$ in a bank account with 4% interest, compounded continuously. (b) gives the same, but compounded $1,000,000$ times a year, which is a very close approximation of continuous compounding.

2. "The more often you compound, the more money you make, so it's worth insisting on an initial investment of $1,000 being compounded every second rather than every minute."

   Critique this statement.

   Not really. While it's true that compounding every second technically results in a slightly higher number, the difference is many decimal points down the line and would not materially affect the amount of money. For example, after 10 years at 5% interest, every minute would get

   $$1000\left(1+\frac{0.05}{365\times 24\times 60}\right)^{365\times 24\times 60\times 10}\approx 1648.72$$

   , and every second

   $$1000\left(1+\frac{0.05}{365\times 24\times 60\times 60}\right)^{365\times 24\times 60\times 60\times 10}\approx 1648.72$$

   So we get the same amount of money.

3. How much money would you have if you invested $1000 at 50% annual interest, compounded twice a year? Don't use a formula here: show computations from scratch.

   25% interest, added twice. After once, we have $\$1,000 * 1.25 = 1\$1,250$. After twice, $\$1,250 * 1.25 = \$1,562.50$.

4. Write an equation whose solution answers each of the following continuous compounding questions:

   (a) Write an equation whose solution gives the initial amount of money P_0 you would need to invest to have $100,000 in 20 years, assuming an account paying continuously compounded 6% annual interest.

   Using $A(t) = P_0e^{rt}$:

   $$100,000 = P_0e^{0.06\times 20}.$$

(b) Write an equation whose solution gives the annual interest rate r (compounded continuously) needed for an initial deposit of \$4,000 to grow to \$10,000 in 15 years.

Using $A(t) = P_0 e^{rt}$ with $P_0 = 4{,}000$, $A = 10{,}000$, and $t = 15$:

$$10{,}000 = 4{,}000e^{15r} \quad \left(\text{or } 2.5 = e^{15r}\right).$$

(c) Write an equation whose solution gives the number of years t it will take for an initial investment of \$2,500 to triple in value in an account paying 4.8% interest compounded continuously.

Tripling \$2,500 gives \$7,500. Using $A(t) = P_0 e^{rt}$:

$$7{,}500 = 2{,}500e^{0.048t} \quad \left(\text{or } 3 = e^{0.048t}\right).$$

5. Suppose that you have the opportunity either get \$10,000 in ten years, or \$5,000 now, given that you can put the money in a bank account with 8% interest, compounded continuously. Which would you prefer? Explain.

If you take the second option, you'd end up with $5000e^{0.08\times 10} \approx \$11,127$. Definitely better.

Exponential Functions

1. Which of the following functions is exponential?

   (a) $f(x) = 3x + 2$
   (b) $f(x) = x^3$
   (c) $f(x) = 2^x$
   (d) $f(x) = e^{x-1}$
   (e) $f(x) = 7^{x^2}$
   (f) $f(x) = |x| + 7$
   (g) $f(x) = x^x$
   (h) $f(x) = e^x + x$

   An exponential function is one in which the variable appears only in the exponent, not as a base or added outside the exponential term. Hence, the exponential functions here are:

   $$f(x) = 2^x \quad \text{and} \quad f(x) = e^{x-1} = e^{-1} \cdot e^x.$$

2. Suppose that $f(x) = P \cdot a^x$, with $a > 0$ and $a \neq 1$. Which of the following are true regardless of the values of P and a (subject to $a > 0$, $a \neq 1$)? For those that are not always true, give values of P and a that make them true.

   (a) $f(x)$ is increasing.
   (b) $f(x)$ is decreasing.
   (c) $f(x)$ is concave up.
   (d) $f(x)$ is concave down.

   None are always true.

   (a) True for (eg) $a > 1$ and $P > 0$, so $f(x) = 2^x$, eg. Also for $P < 0$ and $a < 1$, so $f(x) = -(0.5)^x$, eg
   (b) True for (eg) $a < 1$ and $P > 0$, so $f(x) = 0.5^x$, eg. Also for $P < 0$ and $a > 1$, so $f(x) = -(2^x)$, eg
   (c) True for any a with $P > 0$, so eg $f(x) = 2^x$ or $f(x) = 0.5^x$.
   (d) True for any a with $P < 0$, so eg $f(x) = -(2^x)$ or $f(x) = -(0.5^x)$.

3. Write each of the following functions in the form $P_0 a^t$. Identify the initial value P_0 and the growth/decay factor a in exact form, and provide a decimal approximation for a to 3 decimal places:

   (a) $f(t) = 22e^{0.2t}$

   $$22e^{0.2t} = 22(e^{0.2})^t \approx 22(1.221)^t.$$

   Here $P_0 = 22$ and $a = e^{0.2} \approx 1.221$.

(b) $g(t) = 85e^{-0.05t}$

$$85e^{-0.05t} = 85(e^{-0.05})^t \approx 85(0.951)^t.$$

Here $P_0 = 85$ and $a = e^{-0.05} \approx 0.951$.

(c) $h(t) = 4.5e^{1.2t}$

$$4.5e^{1.2t} = 4.5(e^{1.2})^t \approx 4.5(3.320)^t.$$

Here $P_0 = 4.5$ and $a = e^{1.2} \approx 3.320$.

4. For each of the following scenarios, find an exponential formula of the form $P(t) = P_0a^t$ from the given data and answer the question. Show all your work clearly without using shortcut formulas:

(a) Suppose that a radioactive substance has a half-life of 4 days. If we start with 20 moles, find a formula of the form P_0a^t for the number of moles left at time t days, and use it to find how much is left after 6 days.

The initial quantity is $P_0 = 20$ moles.
After 4 days, half remains, so $10 = 20a^4 \implies a^4 = \frac{1}{2} \implies a = \left(\frac{1}{2}\right)^{1/4} \approx 0.8409$.
The formula is:

$$N(t) = 20(0.8409)^t.$$

After 6 days:

$$N(6) = 20(0.8409)^6 = 20\left(\frac{1}{2}\right)^{6/4} = 20\left(\frac{1}{2}\right)^{1.5} \approx 7.07 \text{ moles.}$$

(b) A culture of bacteria doubles in population every 5 hours. If the culture initially contains 300 bacteria, find a formula of the form $P(t) = P_0a^t$ for the number of bacteria after t hours, and use it to determine the population after 14 hours.

The initial quantity is $P_0 = 300$ bacteria.
After 5 hours, the population doubles to 600, so:

$$600 = 300a^5 \implies a^5 = 2 \implies a = 2^{1/5} \approx 1.1487.$$

The formula is:

$$P(t) = 300(1.1487)^t.$$

After 14 hours:

$$P(14) = 300(1.1487)^{14} = 300(2^{14/5}) = 300(2^{2.8}) \approx 2087 \text{ bacteria.}$$

(c) A prescribed medication has a half-life of 8 hours in the bloodstream. If a patient is administered an initial dose of 80 mg, find a formula of the form $Q(t) = P_0a^t$ for the amount of medication (in mg) remaining in the body after t hours, and use it to find the amount remaining after 20 hours.

The initial amount is $P_0 = 80$ mg.
After 8 hours, half the medication remains:

$$40 = 80a^8 \implies a^8 = \frac{1}{2} \implies a = \left(\frac{1}{2}\right)^{1/8} \approx 0.9170.$$

The formula is:

$$Q(t) = 80(0.9170)^t.$$

After 20 hours:

$$Q(20) = 80(0.9170)^{20} = 80\left(\frac{1}{2}\right)^{20/8} = 80\left(\frac{1}{2}\right)^{2.5} \approx 14.14 \text{ mg}.$$

5. Suppose that $a > 0$. What coordinate do the graphs of all functions of the form $f(x) = a^x$ pass through?

   $f(0) = a^0 = 1$, regardless of the value of a, so all such functions pass through $(0, 1)$.

Logs

1. Find the quadrupling time of a bank account with 6% continuously compounded interest.

   We use the continuous compounding formula: $A = Pe^{rt}$

   We want to find the time t such that $A = 4P$.

$$\begin{aligned} 4P &= Pe^{0.06t} \\ 4 &= e^{0.06t} \\ \ln 4 &= 0.06t \\ t &= \frac{\ln 4}{0.06} \\ t &\approx \frac{1.386294}{0.06} \\ t &\approx 23.10 \text{ years.} \end{aligned}$$

   The quadrupling time is approximately 23.10 years.

2. (a) Find the inverse of $f(x) = 3e^{2x} - 1$.

   Set $y = 3e^{2x} - 1$. Then

$$y + 1 = 3e^{2x} \quad \Rightarrow \quad e^{2x} = \frac{y+1}{3}$$

$$2x = \ln\left(\frac{y+1}{3}\right) \quad \Rightarrow \quad x = \tfrac{1}{2}\ln\left(\frac{y+1}{3}\right).$$

   Hence the inverse function is

$$f^{-1}(x) = \tfrac{1}{2}\ln\left(\frac{x+1}{3}\right).$$

   (b) Find the inverse of $g(x) = \ln(5x - 2)$.

   Let $y = \ln(5x - 2)$. Then

$$e^y = 5x - 2 \quad \Rightarrow \quad x = \frac{e^y + 2}{5}.$$

   Therefore

$$g^{-1}(x) = \frac{e^x + 2}{5},$$

   (c) Find the inverse of $h(x) = 2^{x+3} + 4$.

   Take $y = 2^{x+3} + 4$.

   Then $y - 4 = 2^{x+3}$.

   So $y - 4 = \ln(2^{x+3}) = (x + 3)\ln 2$

   $x + 3 = \frac{\ln(y-4)}{\ln 2}$

   $x = \frac{\ln(y-4)}{\ln 2} - 3$

   So $h^{-1}(x) = \frac{\ln(x-4)}{\ln 2} - 3$ (valid for $x > 4$).

(d) Find the inverse of $p(x) = \frac{1}{4}\ln(x) + 7$.

Let $y = \frac{1}{4}\ln(x) + 7$. Then

$$y - 7 = \tfrac{1}{4}\ln(x) \quad \Rightarrow \quad \ln(x) = 4(y-7)$$

$$x = e^{4(y-7)}.$$

So the inverse is

$$p^{-1}(x) = e^{4(x-7)},$$

(e) Find the inverse of $q(x) = \ln\big(2 - e^{-x}\big)$.

Start with $y = \ln\big(2 - e^{-x}\big)$. Then

$$e^y = 2 - e^{-x} \quad \Rightarrow \quad e^{-x} = 2 - e^y.$$

$$-x = \ln(2 - e^y) \quad \Rightarrow \quad x = -\ln(2 - e^y).$$

Therefore

$$q^{-1}(x) = -\ln\big(2 - e^x\big),$$

3. (a) Solve $5^x = 12$.

Take natural logs of both sides:

$$\ln(5^x) = \ln 12 \quad \Rightarrow \quad x \ln 5 = \ln 12.$$

Hence

$$x = \frac{\ln 12}{\ln 5}.$$

(b) Solve $10^{2x+1} = 7 \cdot 10^{x-2}$.

Divide both sides by 10^{x-2}:

$$10^{(2x+1)-(x-2)} = 7 \quad \Rightarrow \quad 10^{x+3} = 7.$$

Taking log (base 10) gives

$$x + 3 = \log 7 \quad \Rightarrow \quad x = \log 7 - 3.$$

(c) Solve $9^{7-2x} + 6 = e$.

Start with

$$9^{7-2x} + 6 = e.$$

Then

$$9^{7-2x} = e - 6.$$

But $e - 6 < 0$, whereas $9^{7-2x} > 0$ for every real x. Therefore there is no real solution to the equation.

(d) Solve $\log(2x+3) + \log(x-1) = 1$. (Give exact answer and check if your answers are possible!)

Domain requires $2x + 3 > 0$ and $x - 1 > 0$, so $x > 1$. Combine the logs:

$$\log\big((2x+3)(x-1)\big) = 1 \quad \Rightarrow \quad (2x+3)(x-1) = 10.$$

Expand and solve the quadratic:

$$2x^2 + x - 3 = 10 \quad \Rightarrow \quad 2x^2 + x - 13 = 0.$$

Use the quadratic formula:

$$x = \frac{-1 \pm \sqrt{1 + 104}}{4} = \frac{-1 \pm \sqrt{105}}{4}.$$

Only the root greater than 1 is admissible. Since $\frac{-1+\sqrt{105}}{4} > 1$ and $\frac{-1-\sqrt{105}}{4} < 1$, the solution is

$$x = \frac{-1 + \sqrt{105}}{4}.$$

(e) Solve $\log(4x) - \log(x+1) = \frac{1}{2}$. (Give exact answer and check if your answers are possible.)

Domain: $4x > 0$ and $x + 1 > 0$, so $x > 0$. Combine logs:

$$\log\left(\frac{4x}{x+1}\right) = \tfrac{1}{2} \quad \Rightarrow \quad \frac{4x}{x+1} = 10^{1/2} = \sqrt{10}.$$

Solve the resulting linear equation:

$$4x = \sqrt{10}\,(x+1) \quad \Rightarrow \quad 4x = \sqrt{10}\,x + \sqrt{10}$$

$$(4 - \sqrt{10})x = \sqrt{10} \quad \Rightarrow \quad x = \frac{\sqrt{10}}{4 - \sqrt{10}}.$$

This x is positive (so it satisfies the domain restriction), and is the exact solution.

4. Let $x = \log A$ and $y = \log B$. Rewrite each of the following in terms of x and y:

(a) $\log \sqrt{\dfrac{A}{B}}$

$$\log \sqrt{\frac{A}{B}} = \log \left(\frac{A}{B}\right)^{1/2} = \tfrac{1}{2}\big(\log A - \log B\big) = \tfrac{1}{2}(x - y).$$

(b) $\log\left(\dfrac{A^3\sqrt{B}}{10}\right)$

$$\log\left(\frac{A^3\sqrt{B}}{10}\right) = \log A^3 + \log B^{1/2} - \log 10 = 3\log A + \tfrac{1}{2}\log B - 1 = 3x + \tfrac{1}{2}y - 1.$$

(c) $\log\left(\dfrac{\sqrt[3]{A}\,B^{-2}}{A^{-1}}\right)$

First simplify the argument:

$$\frac{\sqrt[3]{A}\,B^{-2}}{A^{-1}} = A^{1/3}B^{-2}A^1 = A^{4/3}B^{-2}.$$

Hence

$$\log\big(A^{4/3}B^{-2}\big) = \tfrac{4}{3}\log A - 2\log B = \tfrac{4}{3}x - 2y.$$

(d) $\dfrac{\log\left(A^2B\right)}{\log\left(\sqrt{AB}\,\right)}$

Evaluate numerator and denominator:

$$\log\left(A^2B\right) = 2\log A + \log B = 2x + y,$$

$$\log\left(\sqrt{AB}\,\right) = \tfrac{1}{2}\log(AB) = \tfrac{1}{2}(\log A + \log B) = \tfrac{1}{2}(x+y).$$

Therefore

$$\frac{\log\left(A^2B\right)}{\log\left(\sqrt{AB}\,\right)} = \frac{2x+y}{\frac{1}{2}(x+y)} = \frac{2(2x+y)}{x+y}.$$

5. Write 9^x in the form 8^{kx}.

$9^x = 8^{kx}$, so $\ln 9^x = \ln 8^{kx}$.

$x \ln 9 = kx \ln 8$, or $k = \frac{\ln 9}{\ln 8} \approx 1.0566$.

So $9^x \approx 8^{1.0566x}$.

Chain Rule

1. Find the derivatives of the following functions:

(a) $y = (2x^3 - x)^5$.

Let $f(x) = x^5$, $g(x) = 2x^3 - x$. Then $f'(x) = 5x^4$, $g'(x) = 6x^2 - 1$. Thus,

$$\frac{dy}{dx} = f'(g(x))g'(x) = 5(2x^3 - x)^4(6x^2 - 1).$$

(b) $y = e^{3x^2+5}$.

Let $f(x) = e^x$ and $g(x) = 3x^2 + 5$. Then $f'(x) = e^x$, $g'(x) = 6x$. So,

$$\frac{dy}{dx} = f'(g(x))g'(x) = e^{3x^2+5}(6x).$$

(c) $y = ((x^2 + 1)^3 + 2)^4$.

(Triple Chain Rule) Let $f(x) = x^4$, $g(x) = x^3+2$, $h(x) = x^2+1$. Then $f'(x) = 4x^3$, $g'(x) = 3x^2$, $h'(x) = 2x$. Therefore,

$$\frac{dy}{dx} = f'(g(h(x)))g'(h(x))h'(x) = 4((x^2 + 1)^3 + 2)^3(3(x^2 + 1)^2)(2x).$$

Simplify:

$$\frac{dy}{dx} = 24x((x^2 + 1)^3 + 2)^3(x^2 + 1)^2.$$

(d) $y = (e^{x^3+3x^2+2x})^2$.

(Triple Chain Rule) Let $f(x) = x^2$, $g(x) = e^x$, $h(x) = x^3 + 3x^2 + 2x$. Then $f'(x) = 2x$, $g'(x) = e^x$, $h'(x) = 3x^2 + 6x + 2$. So,

$$\frac{dy}{dx} = f'(g(h(x)))g'(h(x))h'(x) = 2(e^{x^3+3x^2+2x})(e^{x^3+3x^2+2x})(3x^2 + 6x + 2).$$

Simplify:

$$\frac{dy}{dx} = 2e^{2(x^3+3x^2+2x)}(3x^2 + 6x + 2).$$

(e) $y = (x^4 + 2x^2 + 3)^{1/3}$.

Let $f(x) = x^{1/3}$, $g(x) = x^4 + 2x^2 + 3$. Then $f'(x) = \frac{1}{3}x^{-2/3}$, $g'(x) = 4x^3 + 4x$. Hence,

$$\frac{dy}{dx} = f'(g(x))g'(x) = \frac{1}{3}(x^4 + 2x^2 + 3)^{-2/3}(4x^3 + 4x).$$

(f) $y = e^{(4x^3+1)^{1/2}}$.

(Triple Chain Rule) Let $f(x) = e^x$, $g(x) = x^{1/2}$, $h(x) = 4x^3 + 1$. Then $f'(x) = e^x$, $g'(x) = \frac{1}{2}x^{-1/2}$, $h'(x) = 12x^2$. So,

$$\frac{dy}{dx} = f'(g(h(x)))g'(h(x))h'(x) = e^{(4x^3+1)^{1/2}}\frac{1}{2}(4x^3 + 1)^{-1/2}(12x^2).$$

Simplify:

$$\frac{dy}{dx} = \frac{6x^2e^{(4x^3+1)^{1/2}}}{(4x^3 + 1)^{1/2}}.$$

(g) $y = (5x^4 - 7x^2 + 2)^3$.

Let $f(x) = x^3$, $g(x) = 5x^4 - 7x^2 + 2$. Then $f'(x) = 3x^2$, $g'(x) = 20x^3 - 14x$. Thus,

$$\frac{dy}{dx} = f'(g(x))g'(x) = 3(5x^4 - 7x^2 + 2)^2(20x^3 - 14x).$$

(h) $y = ((x^2 + 1)^4 + 3x)^5$.

(Triple Chain Rule) Let $f(x) = x^5$, $g(x) = x^4 + 3x$, $h(x) = x^2 + 1$. Then $f'(x) = 5x^4$, $g'(x) = 4x^3 + 3$, $h'(x) = 2x$. Therefore,

$$\frac{dy}{dx} = f'(g(h(x)))g'(h(x))h'(x) = 5((x^2 + 1)^4 + 3x)^4(4(x^2 + 1)^3(2x) + 3(2x)).$$

(i) $y = e^{(x^2+1)^3}$.

(Triple Chain Rule) Let $f(x) = e^x$, $g(x) = x^3$, $h(x) = x^2 + 1$. Then $f'(x) = e^x$, $g'(x) = 3x^2$, $h'(x) = 2x$. Hence,

$$\frac{dy}{dx} = f'(g(h(x)))g'(h(x))h'(x) = e^{(x^2+1)^3}(3(x^2 + 1)^2)(2x).$$

Simplify:

$$\frac{dy}{dx} = 6x(x^2 + 1)^2 e^{(x^2+1)^3}.$$

(j) $y = (3x^2 + (x + 1)^3)^4$.

(Triple Chain Rule) Let $f(x) = x^4$, $g(x) = 3x^2 + x^3 + 3x^2 + 3x + 1$, $h(x) = x + 1$. Then $f'(x) = 4x^3$, $g'(x) = 6x + 3(x + 1)^2$, $h'(x) = 1$. Thus,

$$\frac{dy}{dx} = f'(g(h(x)))g'(h(x))h'(x) = 4(3x^2 + (x + 1)^3)^3(6x + 3(x + 1)^2).$$

2. Supose that $f(x)$ is a differentiable function. Differentiate the following. Most involve the triple chain rule.

(a) $f(x)^9$

Let the outer function be $F(x) = x^9$ and the inner be $f(x)$. Then

$$\frac{d}{dx} f(x)^9 = F'(f(x)) \cdot f'(x) = 9f(x)^8 f'(x).$$

(b) $f\big((x^4 + 2)^5\big)$

(Triple Chain Rule)

Let $f(g(h(x)))$ where $g(x) = x^5$, $h(x) = x^4 + 2$. Then

$$\frac{d}{dx} f\big((x^4 + 2)^5\big) = f'\big((x^4 + 2)^5\big) \cdot 5(x^4 + 2)^4 \cdot 4x^3$$

$$= 20x^3(x^4 + 2)^4 f'\big((x^4 + 2)^5\big).$$

(c) $e^{f(x)^2+1}$

(Triple Chain Rule)

Let $F(x) = e^x$, $g(x) = x^2 + 1$. So $F'(x) = e^x$, and $g'(x) = 2x$, giving:

$$\frac{d}{dx}e^{f(x)^2+1} = e^{f(x)^2+1} \cdot (2f(x)f'(x)).$$

Hence

$$\frac{dy}{dx} = 2f(x)f'(x)e^{f(x)^2+1}.$$

(d) $2^{f(x)^3}$

(Triple Chain Rule)

Let $F(x) = 2^x$ and $g(x) = f(x)^3$. Innermost is $f(x)$ Then $F'(x) = 2^x \ln 2$. Thus

$$\frac{d}{dx}2^{f(x)^3} = 2^{f(x)^3} \ln(2) \cdot 3f(x)^2 f'(x).$$

(e) $\big(f(x)^4 + x\big)^8$

(Triple Chain Rule)

Outer $F(x) = x^8$, $g(x) = f(x)^4 + x$. Innermost is $f(x)$ Then

$$\frac{d}{dx}\big(f(x)^4 + x\big)^8 = 8\big(f(x)^4 + x\big)^7 \cdot (4f(x)^3 f'(x) + 1).$$

(f) $f\big((x^2 + 1)^7\big)$

(Triple Chain Rule)

Outermost is $f(x)$. Let $g(x) = x^7$, $h(x) = x^2 + 1$,

$$\frac{dy}{dx} = f'\big((x^2 + 1)^7\big) \cdot 7(x^2 + 1)^6 \cdot 2x = 14x(x^2 + 1)^6 f'\big((x^2 + 1)^7\big).$$

(g) $(10^{f(x)} + 1)^5$

(Triple Chain Rule)

Let $F(x) = x^5$, $G(x) = 10^x + 1$. Innermost is $f(x)$ Then $F'(x) = 5x^4$, $G'(x) = 10^x \ln 10$. Hence

$$\frac{dy}{dx} = 5(10^{f(x)} + 1)^4 (10^{f(x)} \ln 10) f'(x).$$

(h) $f\big(e^{x^3+2x}\big)$

(Triple Chain Rule)

Outermost is $f(x)$. Let $g(x) = e^x$, $h(x) = x^3 + 2x$. Then

$$\frac{d}{dx}f(e^{x^3+2x}) = f'(e^{x^3+2x}) \cdot e^{x^3+2x} \cdot (3x^2 + 2).$$

(i) $(f(x)^5 + 2)^9$

(Triple Chain Rule)

Outer $F(x) = x^9$, middle $G(x) = x^5 + 2$, inner $f(x)$. Then

$$\frac{dy}{dx} = 9(f(x)^5 + 2)^8 \cdot 5f(x)^4 f'(x) = 45f(x)^4 f'(x)(f(x)^5 + 2)^8.$$

(j) $f\left((x^4+3x^2+1)^3\right)$

(Triple Chain Rule)

Outermost is $f(x)$. With $g(x)=x^3$, $h(x)=x^4+3x^2+1$,

$$\frac{dy}{dx} = f'\left((x^4+3x^2+1)^3\right)\cdot 3(x^4+3x^2+1)^2(4x^3+6x).$$

3. Let $f(x)$ and $g(x)$ be differentiable. Find a formula for the derivative of $\frac{f(x)}{(g(x))^2}$. Then ***use it*** to compute the derivative of $\frac{e^x}{x^4}$.

$$\begin{aligned}\frac{d}{dx}\frac{f(x)}{(g(x))^2} &= \frac{d}{dx}f(x)(g(x))^{-2}\\ &= f'(x)(g(x))^{-2} + f(x)\frac{d}{dx}(g(x))^{-2}\end{aligned}$$

$\frac{d}{dx}(g(x))^{-2}$: Inner function $g(x)$, outer $f(x)=x^{-2}$.

So $f'(x)=-2x^{-3}$, giving $\frac{d}{dx}(g(x))^{-2} = -2(g(x))^{-3}g'(x)$.

Plugging back in:

$$\begin{aligned}\frac{d}{dx}\frac{f(x)}{(g(x))^2} &= f'(x)(g(x))^{-2} + f(x)\frac{d}{dx}(g(x))^{-2}\\ &= f'(x)(g(x))^{-2} + f(x)\times -2(g(x))^{-3}g'(x)\\ &= \frac{f'(x)}{g(x)^2} - \frac{2f(x)g'(x)}{g(x)^3}\\ &= \frac{f'(x)g(x) - 2f(x)g'(x)}{g(x)^3}\end{aligned}$$

Let $f(x)=e^x$ and $g(x)=x^2$, then $\frac{f(x)}{g(x)^2} = \frac{e^x}{x^4}$. Also, $f'(x)=e^x$ and $g'(x)=2x$.

We get:

$$\begin{aligned}\frac{d}{dx}\frac{e^x}{x^4} &= \frac{e^x\cdot x^2 - 2e^x\cdot 2x}{x^6}\\ &= \frac{xe^x(x-4)}{x^6}\\ &= \frac{e^x(x-4)}{x^5}\end{aligned}$$

4. For each of the following real-world composite scenarios, interpret the functions and apply the Chain Rule:

   (a) Suppose that the depth of a lake (in feet) at a distance of d feet from its center is given by a function $f(d)$. Suppose also that the density of fish (in fish/ft^3) in an area of the lake with depth D feet is given by a function $p(D)$.

(i) Explain what the composite function $F(d) = p(f(d))$ means in practical terms.

(ii) Explain why the expression $f(p(d))$ does not make physical sense.

(iii) What are the units of $f'(d)$, $p'(D)$, and $F'(d)$?

(iv) Suppose that at a distance of 1,000 feet from the center of the lake, the depth is 20 feet and is getting shallower at a rate of 0.1 ft of depth for each additional foot from the center. Furthermore, when the depth is 20 ft, each additional foot of depth increases the density of fish by 100 fish/ft^3. Compute the rate at which the density of fish changes per foot of distance from the center when you are 1,000 feet from the center. Give units.

(i) $F(d)$ gives the density of fish (in fish/ft^3) at a distance of d feet from the center of the lake.

(ii) $p(d)$ takes depth (in feet) and outputs density (fish/ft^3). But f expects an input of distance from the center in feet. Density cannot be plugged into a function that requires distance.

(iii) $f'(d)$: feet of depth per foot of distance from center (ft/ft); $p'(D)$: (fish/ft^3) per foot of depth; $F'(d)$: (fish/ft^3) per foot of distance from center.

(iv) We are given $f(1000) = 20$ ft, $f'(1000) = -0.1$ ft/ft (negative since depth is decreasing), and $p'(20) = 100$ (fish/ft^3)/ft. By the Chain Rule:

$$F'(1000) = p'(f(1000)) \cdot f'(1000) = p'(20) \cdot f'(1000) = 100 \times (-0.1) = -10 \ \frac{\text{fish/ft}^3}{\text{ft}}.$$

Thus, the density of fish decreases by 10 fish/ft^3 for every foot further from the center.

(b) A research balloon ascends into the atmosphere. Let $h(t)$ be the altitude of the balloon (in kilometers) t hours after launch, and let $T(h)$ be the atmospheric temperature (in °C) at altitude h kilometers above sea level.

(i) Explain what the composite function $C(t) = T(h(t))$ means in practical terms.

(ii) Explain why the expression $h(T(t))$ does not make physical sense.

(iii) State the units of $h'(t)$, $T'(h)$, and $C'(t)$.

(iv) Suppose that 2 hours after launch, the balloon is at an altitude of 6 km and is climbing at a rate of 1.5 km/hr. At an altitude of 6 km, atmospheric temperature decreases at a rate of 6.5°C per kilometer of altitude increase. Compute $C'(2)$, the rate of change of the temperature experienced by the balloon with respect to time at $t = 2$ hours. Include units.

(i) $C(t)$ gives the atmospheric temperature (in °C) experienced by the balloon t hours after launch.

(ii) T takes altitude in kilometers as its input, not time t in hours. Furthermore, the output of T is temperature in °C, whereas h requires an input of time, so the units and physical meanings do not match.

(iii) $h'(t)$: kilometers per hour (km/hr); $T'(h)$: °C per kilometer (°C/km); $C'(t)$: °C per hour (°C/hr).

(iv) We are given $h(2) = 6$ km, $h'(2) = 1.5$ km/hr, and $T'(6) = -6.5$°C/km (negative since temperature is decreasing with altitude). By the Chain Rule:

$$C'(2) = T'(h(2)) \cdot h'(2) = T'(6) \cdot h'(2) = (-6.5) \times (1.5) = -9.75\text{°C/hr}.$$

The temperature around the balloon is dropping at a rate of 9.75°C per hour.

(c) An electric car's performance is tested. Let $s(t)$ be the speed of the car (in miles per hour) t minutes after entering a highway, and let $E(s)$ be the vehicle's energy consumption rate (in watt-hours per mile, Wh/mi) when traveling at speed s mph.

(i) Explain what the composite function $P(t) = E(s(t))$ represents in practical terms.

(ii) Explain why the expression $s(E(t))$ does not make physical sense.

(iii) State the units of $s'(t)$, $E'(s)$, and $P'(t)$.

(iv) Suppose that 10 minutes into the drive, the car is travelling at 65 mph and accelerating at a rate of 1.2 mph per minute. At a speed of 65 mph, aerodynamic drag causes the energy consumption rate to increase by 4 Wh/mi for every 1 mph increase in speed. Compute $P'(10)$ and include units.

(i) $P(t)$ represents the vehicle's instantaneous energy consumption rate (in Wh/mi) t minutes after entering the highway.

(ii) E requires speed in mph as its input, not time in minutes. Additionally, s requires time as input rather than an energy consumption rate.

(iii) $s'(t)$: mph per minute; $E'(s)$: (Wh/mi)/mph; $P'(t)$: (Wh/mi)/min.

(iv) We are given $s(10) = 65$ mph, $s'(10) = 1.2$ mph/min, and $E'(65) = 4$ (Wh/mi)/mph. By the Chain Rule:

$$P'(10) = E'(s(10)) \cdot s'(10) = E'(65) \cdot s'(10) = 4 \times 1.2 = 4.8 \ \frac{\text{Wh/mi}}{\text{min}}.$$

The car's energy consumption rate is increasing at a rate of 4.8 Wh/mi per minute.

Differentiating Exponentials

Note: some of the chain rule problems above cover derivatives of exponentials.

1. Fill in the blanks: Let $a > 0$. Then the derivative of $f(x) = a^x$ is directly proportional to $f(x)$ with constant of proportionality $\ln a$.

2. Suppose that $e^{g(x)} = x$. By differentiating both side of this equation, show that $g'(x) = \frac{1}{x}$.

   Chain rule. Inner function is $g(x)$, outer is $f(x) = e^x$, whose derivative is $f'(x) = e^x$.

   So $f'(g(x)) = e^{g(x)}$.

   $\frac{d}{dx}e^{g(x)} = e^{g(x)}g'(x)$ and $\frac{d}{dx}x = 1$.

   So $e^{g(x)}g'(x) = 1$. Since $e^{g(x)} = x$, we get $xg'(x) = 1$, or $g'(x) = \frac{1}{x}$.

3. Find the slope of the tangent line to each of the following functions at the given value of x:

   (a) $f(x) = 2^x + e^x + e^2 + 2^e + x^2 + x^e$ at $x = 1$.

   Recall that e^2 and 2^e are constants, so their derivatives are 0. Using the exponential and power rules:

   $$f'(x) = (\ln 2) \cdot 2^x + e^x + 0 + 0 + 2x + ex^{e-1}.$$

   Evaluating at $x = 1$:

   $$f'(1) = (\ln 2) \cdot 2^1 + e^1 + 2(1) + e(1)^{e-1} = 2\ln 2 + e + 2 + e = 2\ln 2 + 2e + 2.$$

   (b) $g(x) = 3^x - 2e^x + x^3 + 3^e$ at $x = 0$.

   Note that 3^e is a constant. Differentiating:

   $$g'(x) = (\ln 3) \cdot 3^x - 2e^x + 3x^2 + 0.$$

   Evaluating at $x = 0$:

   $$g'(0) = (\ln 3) \cdot 3^0 - 2e^0 + 3(0)^2 = \ln 3 - 2.$$

   (c) $h(x) = 5 \cdot 2^x + 4e^x - 3x^2 + \pi^2$ at $x = 1$.

   Note that π^2 is a constant. Differentiating:

   $$h'(x) = 5(\ln 2) \cdot 2^x + 4e^x - 6x + 0.$$

   Evaluating at $x = 1$:

   $$h'(1) = 5(\ln 2)(2) + 4e - 6 = 10\ln 2 + 4e - 6.$$

   (d) $p(x) = 10^x - x^{10} + 10^{10} + e^x$ at $x = 1$.

   Note that 10^{10} is a constant. Differentiating:

   $$p'(x) = (\ln 10) \cdot 10^x - 10x^9 + 0 + e^x.$$

   Evaluating at $x = 1$:

   $$p'(1) = 10\ln 10 - 10(1) + e = 10\ln 10 - 10 + e.$$

(e) $q(x) = 4e^x + 5^x + x^5 + 5^5$ at $x = 0$.

Note that 5^5 is a constant. Differentiating:

$$q'(x) = 4e^x + (\ln 5) \cdot 5^x + 5x^4 + 0.$$

Evaluating at $x = 0$:

$$q'(0) = 4(1) + (\ln 5)(1) + 0 = 4 + \ln 5.$$

(f) $r(x) = 2 \cdot 3^x + e^x - e \cdot x^{e-1} + e^\pi$ at $x = 1$.

Note that e^π is a constant. Differentiating:

$$r'(x) = 2(\ln 3) \cdot 3^x + e^x - e(e-1)x^{e-2} + 0.$$

Evaluating at $x = 1$:

$$r'(1) = 6 \ln 3 + e - e(e-1) = 6 \ln 3 + 2e - e^2.$$

4. Find a function of the form $f(x) = a^x$, so that $f'(x) = -f(x)$.

$\frac{d}{dx} a^x = \ln a \cdot a^x$.

This must equal $-a^x$, so we get $\ln a \cdot a^x = -a^x$, or $\ln a = -1$, or $a = e^{-1}$.

5. What is the exact value of $\lim_{h \to 0} \dfrac{7^h - 1}{h}$? Explain.

$\lim_{h \to 0} \dfrac{7^h - 1}{h} = \ln 7$.

We know that $\frac{d}{dx} 7^x = (\ln 7) 7^x$. Computing the same, we get:

$$\begin{aligned} \frac{d}{dx} 7^x &= \lim_{h \to 0} \frac{7^{x+h} - 7^x}{h} \\ &= \lim_{h \to 0} \frac{7^x 7^h - 7^x}{h} \\ &= \lim_{h \to 0} \frac{7^x (7^h - 1)}{h} \\ &= 7^x \lim_{h \to 0} \frac{7^h - 1}{h} \end{aligned}$$

Therefore $7^x \lim_{h \to 0} \dfrac{7^h - 1}{h} = (\ln 7) 7^x$. Divide by 7^x to get the result.

6. Find the derivative of each of the following exponential functions. Simplify your answers where appropriate:

(a) $f(x) = 5e^x - 4 \cdot 3^x$

Using the exponential rules $\frac{d}{dx} e^x = e^x$ and $\frac{d}{dx} a^x = (\ln a) a^x$:

$$f'(x) = 5e^x - 4(\ln 3) \cdot 3^x.$$

(b) $g(x) = e^{5x-2}$

Using the Chain Rule with outer e^u and inner $u = 5x - 2$:

$$g'(x) = e^{5x-2} \cdot \frac{d}{dx}(5x-2) = 5e^{5x-2}.$$

(c) $h(x) = 7^{3x}$

Using the general exponential rule and Chain Rule:

$$h'(x) = (\ln 7) \cdot 7^{3x} \cdot \frac{d}{dx}(3x) = 3(\ln 7) \cdot 7^{3x}.$$

(d) $k(x) = x^4 e^x$

Using the Product Rule:

$$k'(x) = \frac{d}{dx}(x^4) \cdot e^x + x^4 \cdot \frac{d}{dx}(e^x) = 4x^3 e^x + x^4 e^x = x^3(4+x)e^x.$$

(e) $m(x) = \frac{e^x}{x^2+3}$

Using the Quotient Rule:

$$m'(x) = \frac{(x^2+3)\frac{d}{dx}(e^x) - e^x \frac{d}{dx}(x^2+3)}{(x^2+3)^2} = \frac{(x^2+3)e^x - 2xe^x}{(x^2+3)^2} = \frac{(x^2-2x+3)e^x}{(x^2+3)^2}.$$

(f) $n(x) = 3 \cdot 10^{x^2}$

Using the Chain Rule:

$$n'(x) = 3(\ln 10) \cdot 10^{x^2} \cdot \frac{d}{dx}(x^2) = 3(\ln 10) \cdot 10^{x^2} \cdot (2x) = 6x(\ln 10) \cdot 10^{x^2}.$$

(g) $p(x) = e^{-x^2/2}$

Using the Chain Rule with $u = -x^2/2$:

$$p'(x) = e^{-x^2/2} \cdot \frac{d}{dx}\left(-\frac{1}{2}x^2\right) = e^{-x^2/2} \cdot (-x) = -xe^{-x^2/2}.$$

(h) $q(x) = (2^x + 5)^3$

Using the Chain Rule (Power Rule followed by exponential derivative):

$$q'(x) = 3(2^x+5)^2 \cdot \frac{d}{dx}(2^x+5) = 3(2^x+5)^2 \cdot ((\ln 2)2^x) = 3(\ln 2)2^x(2^x+5)^2.$$

(i) $r(x) = \frac{3e^x+2}{e^x-1}$

Using the Quotient Rule:

$$\begin{aligned} r'(x) &= \frac{(e^x-1)\frac{d}{dx}[3e^x+2] - (3e^x+2)\frac{d}{dx}[e^x-1]}{(e^x-1)^2} \\ &= \frac{(e^x-1)(3e^x) - (3e^x+2)(e^x)}{(e^x-1)^2} \\ &= \frac{3e^{2x} - 3e^x - 3e^{2x} - 2e^x}{(e^x-1)^2} = \frac{-5e^x}{(e^x-1)^2}. \end{aligned}$$

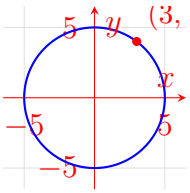
(j) $s(x) = (2x-1)e^{3x}$

Using the Product Rule and Chain Rule:

$$\begin{aligned} s'(x) &= \frac{d}{dx}[2x-1]\cdot e^{3x} + (2x-1)\cdot \frac{d}{dx}[e^{3x}] \\ &= 2e^{3x} + (2x-1)(3e^{3x}) \\ &= (2+6x-3)e^{3x} = (6x-1)e^{3x}. \end{aligned}$$

Implicit Differentiation

1. Consider the circle defined by the equation $x^2 + y^2 = 25$:



   (a) Find $\frac{dy}{dx}$ using implicit differentiation.
   Differentiating both sides with respect to x:

$$2x + 2y\frac{dy}{dx} = 0 \implies 2y\frac{dy}{dx} = -2x \implies \frac{dy}{dx} = -\frac{x}{y}.$$

   (b) Find the equation of the tangent line to the circle at the point $(3, 4)$.
   At $(3, 4)$, the slope is $m = -\frac{3}{4}$. Using point-slope form:

$$y - 4 = -\frac{3}{4}(x - 3) \implies y = -\frac{3}{4}x + \frac{25}{4}.$$

   (c) Find all points (x, y) on the circle where the tangent line is horizontal.
   A horizontal tangent line occurs when $\frac{dy}{dx} = 0$, so $-x = 0 \implies x = 0$.
   Substituting into $x^2 + y^2 = 25$:

$$0^2 + y^2 = 25 \implies y = \pm 5.$$

   Since the denominator $y \neq 0$ at these points, the horizontal tangent lines occur at $(0, 5)$ and $(0, -5)$.

   (d) Find all points (x, y) on the circle where the tangent line is vertical.
   A vertical tangent line occurs when the denominator of $\frac{dy}{dx}$ is zero (and the numerator is non-zero):

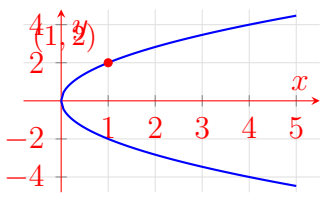
$$y = 0.$$

   Substituting into $x^2 + y^2 = 25$:

$$x^2 + 0^2 = 25 \implies x = \pm 5.$$

   Since the numerator $-x = \mp 5 \neq 0$, the vertical tangent lines occur at $(5, 0)$ and $(-5, 0)$.

2. Consider the parabola opening to the right defined by the equation $y^2 = 4x$:



(a) Find $\frac{dy}{dx}$ using implicit differentiation.

Differentiating both sides with respect to x:

$$2y\frac{dy}{dx} = 4 \implies \frac{dy}{dx} = \frac{4}{2y} = \frac{2}{y}.$$

(b) Find the equation of the tangent line to the parabola at the point $(1, 2)$.

At $(1, 2)$, the slope is $m = \frac{2}{2} = 1$. Using point-slope form:

$$y - 2 = 1(x - 1) \implies y = x + 1.$$

(c) Explain why this parabola has no horizontal tangent lines.

A horizontal tangent requires $\frac{dy}{dx} = 0$. Since $\frac{dy}{dx} = \frac{2}{y}$ has a constant non-zero numerator ($2 \neq 0$), $\frac{dy}{dx}$ can never equal 0. Thus, there are no horizontal tangent lines.

(d) Find the point (x, y) on the parabola where the tangent line is vertical.

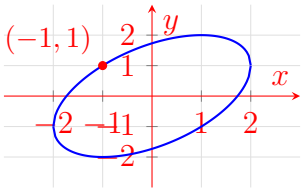
A vertical tangent occurs when the denominator is zero:

$$y = 0.$$

Substituting $y = 0$ into $y^2 = 4x$ gives $0 = 4x \implies x = 0$.

Thus, the vertical tangent line occurs at the vertex $(0, 0)$. (The vertical tangent line is the y-axis, $x = 0$.)

3. Consider the rotated ellipse defined by the equation $x^2 - xy + y^2 = 3$:



(a) Verify that the point $(-1, 1)$ lies on the curve, and compute $\frac{dy}{dx}$ in terms of x and y.

Plugging $(-1, 1)$ into the equation:

$$(-1)^2 - (-1)(1) + 1^2 = 1 + 1 + 1 = 3.$$

Since both sides equal 3, $(-1, 1)$ lies on the curve.

Differentiating both sides with respect to x, using the Product Rule on $-xy$:

$$2x - \left(y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0 \implies 2x - y + (2y - x)\frac{dy}{dx} = 0.$$

Solving for $\frac{dy}{dx}$:

$$\frac{dy}{dx}(2y - x) = y - 2x \implies \frac{dy}{dx} = \frac{y - 2x}{2y - x}.$$

(b) Find the equation of the tangent line to the curve at the point $(-1, 1)$.

Evaluating $\frac{dy}{dx}$ at $(-1, 1)$:

$$m = \frac{1 - 2(-1)}{2(1) - (-1)} = \frac{1+2}{2+1} = \frac{3}{3} = 1.$$

Using point-slope form:

$$y - 1 = 1(x - (-1)) \implies y - 1 = x + 1 \implies y = x + 2.$$

(c) Find all points (x, y) on the curve where the tangent line is horizontal.

A horizontal tangent occurs when the numerator of $\frac{dy}{dx}$ is zero:

$$y - 2x = 0 \implies y = 2x.$$

Substituting $y = 2x$ into the curve equation $x^2 - xy + y^2 = 3$:

$$x^2 - x(2x) + (2x)^2 = 3 \implies 3x^2 = 3 \implies x^2 = 1 \implies x = \pm 1.$$

- If $x = 1$: $y = 2(1) = 2$, giving $(1, 2)$.
- If $x = -1$: $y = 2(-1) = -2$, giving $(-1, -2)$.

Checking denominators: at $(1, 2)$, $2y - x = 4 - 1 = 3 \neq 0$; at $(-1, -2)$, $2y - x = -4 - (-1) = -3 \neq 0$. Thus, horizontal tangent lines occur at $(1, 2)$ and $(-1, -2)$.

(d) Find all points (x, y) on the curve where the tangent line is vertical.

A vertical tangent occurs when the denominator is zero:

$$2y - x = 0 \implies x = 2y.$$

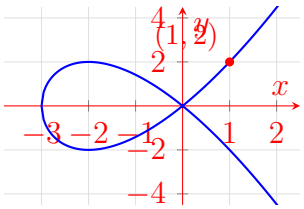
Substituting $x = 2y$ into the curve equation $x^2 - xy + y^2 = 3$:

$$(2y)^2 - (2y)y + y^2 = 3 \implies 3y^2 = 3 \implies y^2 = 1 \implies y = \pm 1.$$

- If $y = 1$: $x = 2(1) = 2$, giving $(2, 1)$.
- If $y = -1$: $x = 2(-1) = -2$, giving $(-2, -1)$.

Checking numerators: at $(2, 1)$, $y - 2x = 1 - 4 = -3 \neq 0$; at $(-2, -1)$, $y - 2x = -1 - (-4) = 3 \neq 0$. Thus, vertical tangent lines occur at $(2, 1)$ and $(-2, -1)$.

4. Consider the nodal cubic curve (also known as the Tschirnhausen cubic) defined by the equation $y^2 = x^3 + 3x^2$:



(a) Find $\frac{dy}{dx}$ using implicit differentiation.

Differentiating both sides with respect to x:

$$2y\frac{dy}{dx} = 3x^2 + 6x \implies \frac{dy}{dx} = \frac{3x^2 + 6x}{2y} = \frac{3x(x+2)}{2y}.$$

(b) Find the equation of the tangent line to the curve at the point $(1, 2)$.

Evaluating $\frac{dy}{dx}$ at $(1, 2)$:

$$m = \frac{3(1)^2 + 6(1)}{2(2)} = \frac{9}{4}.$$

Using point-slope form:

$$y - 2 = \frac{9}{4}(x - 1) \implies y = \frac{9}{4}x - \frac{9}{4} + 2 \implies y = \frac{9}{4}x - \frac{1}{4}.$$

(c) Find all points (x, y) on the curve where the tangent line is horizontal. (Note: at $(0, 0)$, $\frac{dy}{dx}$ is indeterminate $\frac{0}{0}$ because the curve crosses itself.)

A horizontal tangent line occurs when the numerator of $\frac{dy}{dx}$ is zero and the denominator is non-zero:

$$3x(x + 2) = 0 \implies x = 0 \quad \text{or} \quad x = -2.$$

- For $x = -2$: $y^2 = (-2)^3 + 3(-2)^2 = -8 + 12 = 4 \implies y = \pm 2$. At both $(-2, 2)$ and $(-2, -2)$, the denominator $2y = \pm 4 \neq 0$.
- For $x = 0$: $y^2 = 0 \implies y = 0$. At $(0, 0)$, both numerator and denominator vanish $(\frac{0}{0})$, reflecting the fact that the curve crosses itself at the origin with two distinct tangent lines (slopes $\pm\sqrt{3}$).

Thus, horizontal tangent lines occur at $(-2, 2)$ and $(-2, -2)$.

(d) Find all points (x, y) on the curve where the tangent line is vertical.

A vertical tangent occurs when the denominator is zero and the numerator is non-zero:

$$2y = 0 \implies y = 0.$$

Substituting $y = 0$ into the curve equation $y^2 = x^3 + 3x^2$:

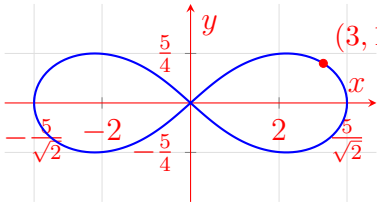
$$0 = x^2(x + 3) \implies x = 0 \quad \text{or} \quad x = -3.$$

- At $x = 0$, $(0, 0)$ gives $\frac{0}{0}$ as discussed in part (c).
- At $x = -3$, the numerator is $3(-3)^2 + 6(-3) = 27 - 18 = 9 \neq 0$.

Thus, a vertical tangent line occurs at the left loop boundary $(-3, 0)$ (tangent line $x = -3$).

5. **[Challenge problem]** Consider the lemniscate of Bernoulli defined by the equation:

$$2(x^2 + y^2)^2 = 25(x^2 - y^2)$$



(a) Verify that the point $(3, 1)$ lies on the curve, and use implicit differentiation to show that:

$$\frac{dy}{dx} = \frac{x(25 - 4(x^2 + y^2))}{y(25 + 4(x^2 + y^2))}.$$

Plugging $(3, 1)$ into the equation:

$$\text{LHS} = 2(3^2 + 1^2)^2 = 2(10)^2 = 200, \quad \text{RHS} = 25(3^2 - 1^2) = 25(8) = 200.$$

Since LHS = RHS, the point $(3,1)$ lies on the curve.
Now differentiate both sides with respect to x, applying the Chain Rule to $(x^2+y^2)^2$:

$$4(x^2+y^2)\left(2x+2y\frac{dy}{dx}\right) = 25\left(2x-2y\frac{dy}{dx}\right).$$

Divide both sides by 2:

$$4(x^2+y^2)\left(x+y\frac{dy}{dx}\right) = 25\left(x-y\frac{dy}{dx}\right).$$

Expanding both sides:

$$4x(x^2+y^2)+4y(x^2+y^2)\frac{dy}{dx} = 25x-25y\frac{dy}{dx}.$$

Grouping terms with $\frac{dy}{dx}$:

$$\frac{dy}{dx}\Big(4y(x^2+y^2)+25y\Big) = 25x-4x(x^2+y^2).$$

Factoring y on the left and x on the right:

$$\frac{dy}{dx}\cdot y\Big(25+4(x^2+y^2)\Big) = x\Big(25-4(x^2+y^2)\Big) \implies \frac{dy}{dx} = \frac{x(25-4(x^2+y^2))}{y(25+4(x^2+y^2))}.$$

(b) Find the equation of the tangent line to the lemniscate at the point $(3,1)$.
At $(3,1)$, we have $x^2+y^2=10$. Evaluating $\frac{dy}{dx}$:

$$m = \frac{3(25-4(10))}{1(25+4(10))} = \frac{3(25-40)}{25+40} = \frac{3(-15)}{65} = -\frac{45}{65} = -\frac{9}{13}.$$

Using point-slope form:

$$y-1 = -\frac{9}{13}(x-3) \implies y = -\frac{9}{13}x+\frac{27}{13}+1 \implies y = -\frac{9}{13}x+\frac{40}{13}.$$

(c) Find the coordinates (x,y) of all points on the lemniscate where the tangent line is horizontal.
A horizontal tangent occurs when the numerator of $\frac{dy}{dx}$ is zero:

$$x(25-4(x^2+y^2)) = 0.$$

- If $x=0$, then $2(y^2)^2 = 25(-y^2) \implies 2y^4 = -25y^2$, which gives only $y=0$. At $(0,0)$, $\frac{dy}{dx} = \frac{0}{0}$ (the origin is the self-intersection of the two loops).
- If $25-4(x^2+y^2)=0$, then $x^2+y^2 = \frac{25}{4}$. Substitute this into the original curve equation $2(x^2+y^2)^2 = 25(x^2-y^2)$:

$$2\left(\frac{25}{4}\right)^2 = 25(x^2-y^2) \implies \frac{625}{8} = 25(x^2-y^2) \implies x^2-y^2 = \frac{25}{8}.$$

We now have the system:

$$\begin{cases} x^2+y^2 = \frac{25}{4} = \frac{50}{8} \\ x^2-y^2 = \frac{25}{8} \end{cases}$$

Adding the equations gives $2x^2 = \frac{75}{8} \implies x^2 = \frac{75}{16} \implies x = \pm\frac{5\sqrt{3}}{4}$.
Subtracting the equations gives $2y^2 = \frac{25}{8} \implies y^2 = \frac{25}{16} \implies y = \pm\frac{5}{4}$.

Checking the denominator: for $y = \pm\frac{5}{4}$, the denominator is non-zero. Thus, horizontal tangent lines occur at the four peak points of the figure-eight:

$$\left(\frac{5\sqrt{3}}{4}, \pm\frac{5}{4}\right) \quad \text{and} \quad \left(-\frac{5\sqrt{3}}{4}, \pm\frac{5}{4}\right).$$

(d) Find the coordinates (x, y) of all points on the lemniscate where the tangent line is vertical.

A vertical tangent occurs when the denominator of $\frac{dy}{dx}$ is zero:

$$y(25 + 4(x^2 + y^2)) = 0.$$

Since $25 + 4(x^2 + y^2) \geq 25 > 0$ for all real (x, y), this requires:

$$y = 0.$$

Substituting $y = 0$ into the curve equation $2(x^2 + y^2)^2 = 25(x^2 - y^2)$:

$$2(x^2)^2 = 25(x^2) \implies 2x^4 = 25x^2 \implies x^2(2x^2 - 25) = 0.$$

- $x = 0$ gives $(0, 0)$ where $\frac{dy}{dx} = \frac{0}{0}$.
- $2x^2 = 25 \implies x^2 = \frac{25}{2} \implies x = \pm\frac{5}{\sqrt{2}} = \pm\frac{5\sqrt{2}}{2}$.

At these points, the numerator is $x(25 - 4(25/2)) = x(25 - 50) = -25x \neq 0$. Thus, vertical tangent lines occur at the two outer tips of the lemniscate:

$$\left(\frac{5\sqrt{2}}{2}, 0\right) \quad \text{and} \quad \left(-\frac{5\sqrt{2}}{2}, 0\right).$$